NUS National University of Singapore
Department of Statistics and Data Science
Faculty of Science

DSS5201 Data Visualisation

Week 8 Principles of Visualisation

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Semester 2 AY25/26

Last Week: Tidy Data / Relational Data

- Tidy Data
  - Reshaping: Wide to Long
  - Reshaping: Long to Wide
  - Splitting a Column
- Relational Data
  - Relations
  - Joins
  - Row Filtering
  - Concatenation

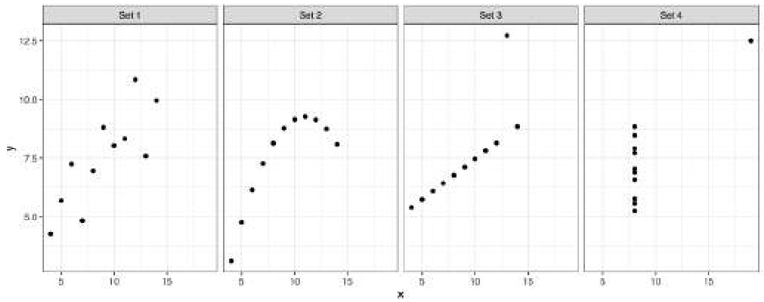
This Week: Principles of Visualisation

- Principles of Visualisation
- Graphical Excellence
- Best Practices & References

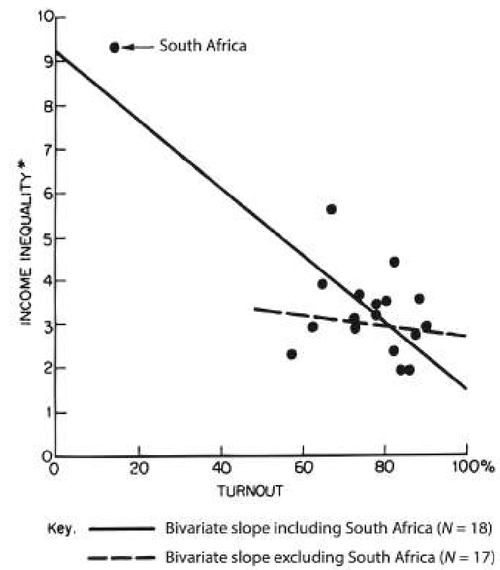
Prologue: Why "Look" at Data?

… instead of relying on tables or numerical summaries

x1	x2	x3	x4	y1	y2	y3	y4
10	10	10	8	8.04	9.14	7.46	6.58
8	8	8	8	6.95	8.14	6.77	5.76
13	13	13	8	7.58	8.74	12.74	7.71
9	9	9	8	8.81	8.77	7.11	8.84
11	11	11	8	8.33	9.26	7.81	8.47
14	14	14	8	9.96	8.10	8.84	7.04
6	6	6	8	7.24	6.13	6.08	5.25
4	4	4	19	4.26	3.10	5.39	12.50



Anscombe (1973)



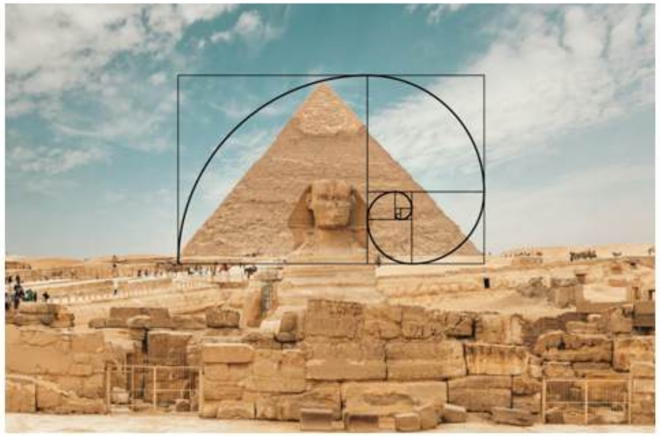
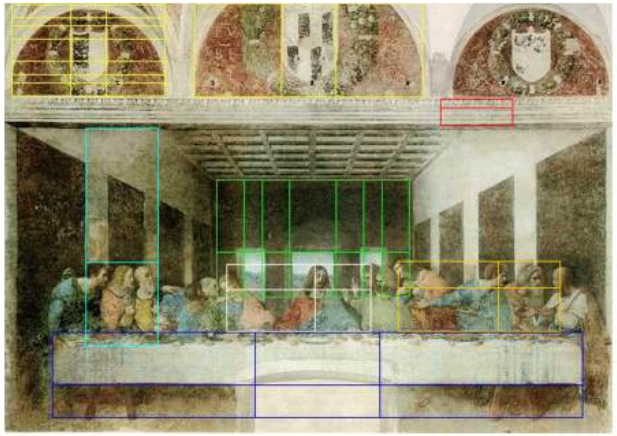
Jackman (1980)

Principles of Visualisation

Learning to Visualise

Convey information through visualisation

Source: Bloomberg, Straits Times, Land Transport Authority, Urban Redevelopment Authority

Despite the inherent subjectivity of beauty, we often repond positively to certain recurring visual pattens that feel balanced and harmonious.

- Symmetry
- Proportion (e.g., the Golden Ratio $1 : \frac{1+\sqrt{5}}{2}$)
- Alignment and Consistency

Subjectivity notwithstanding, there are some techniques and principles we can follow in order to make our visualisations more effective.

Introduction

This week, we explore **key principles of effective data visualisation.**

- This is by no means an exhaustive list.
- You might not agree with some/most/all of them.
- But they are starting points for you to keep in mind when designing visualisations.

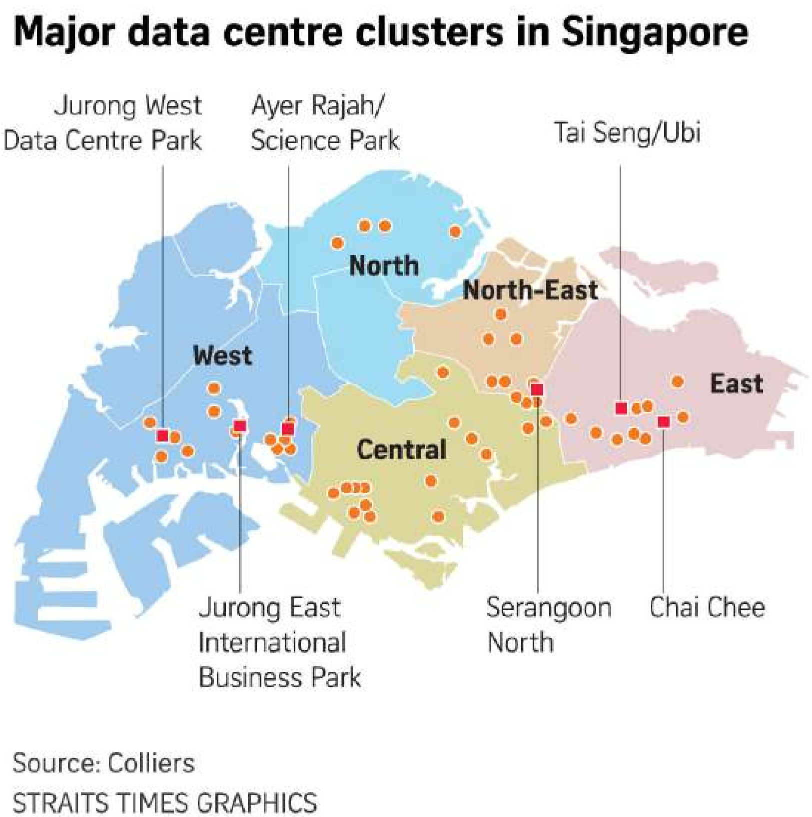
We will

- Show some good graphics.
- Show some bad graphics.
- Identify core principles behind effective data visualisation design.

Types of Graphical Displays

1. Data Maps

Overlay data onto geographic locations to study clusters, hotspots, etc.

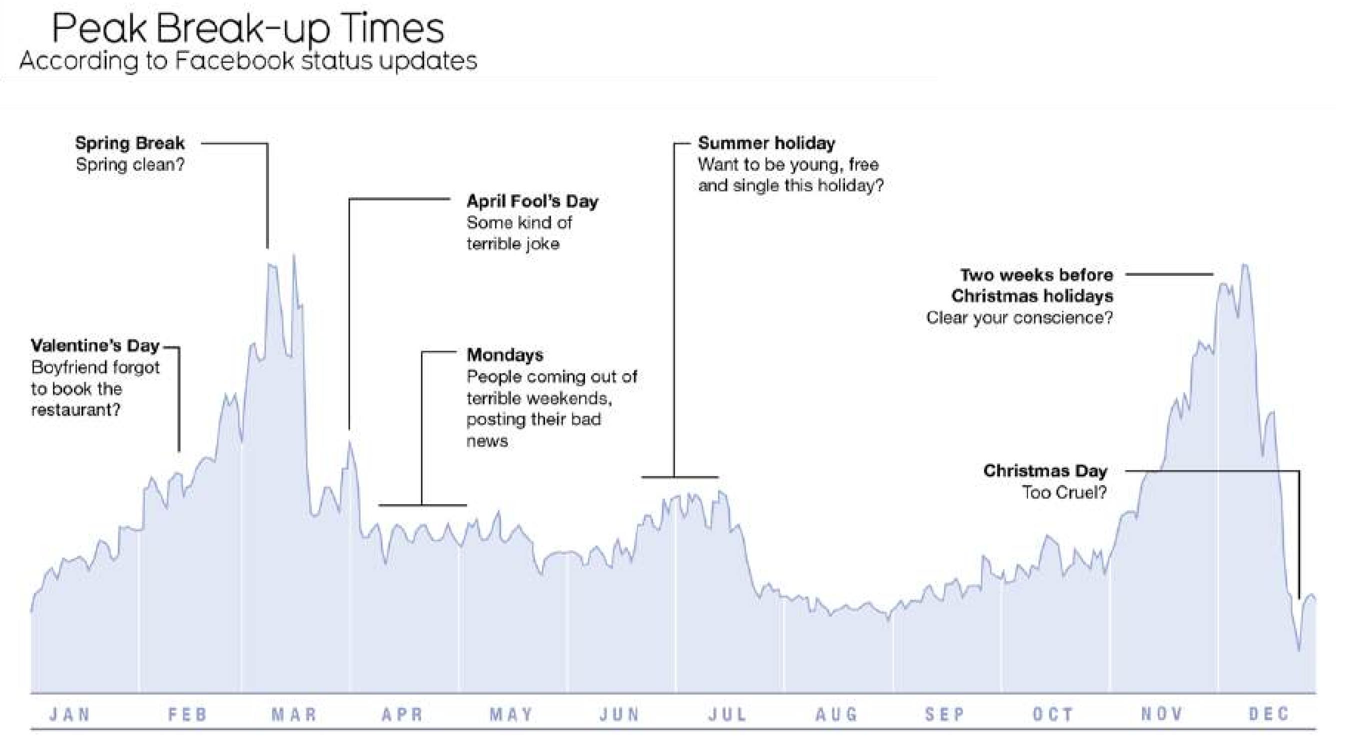


Source: The Straits Times, 11 Jul 2022

Types of Graphical Displays

2. Time Series

Probably the most frequently used form of graphic design.

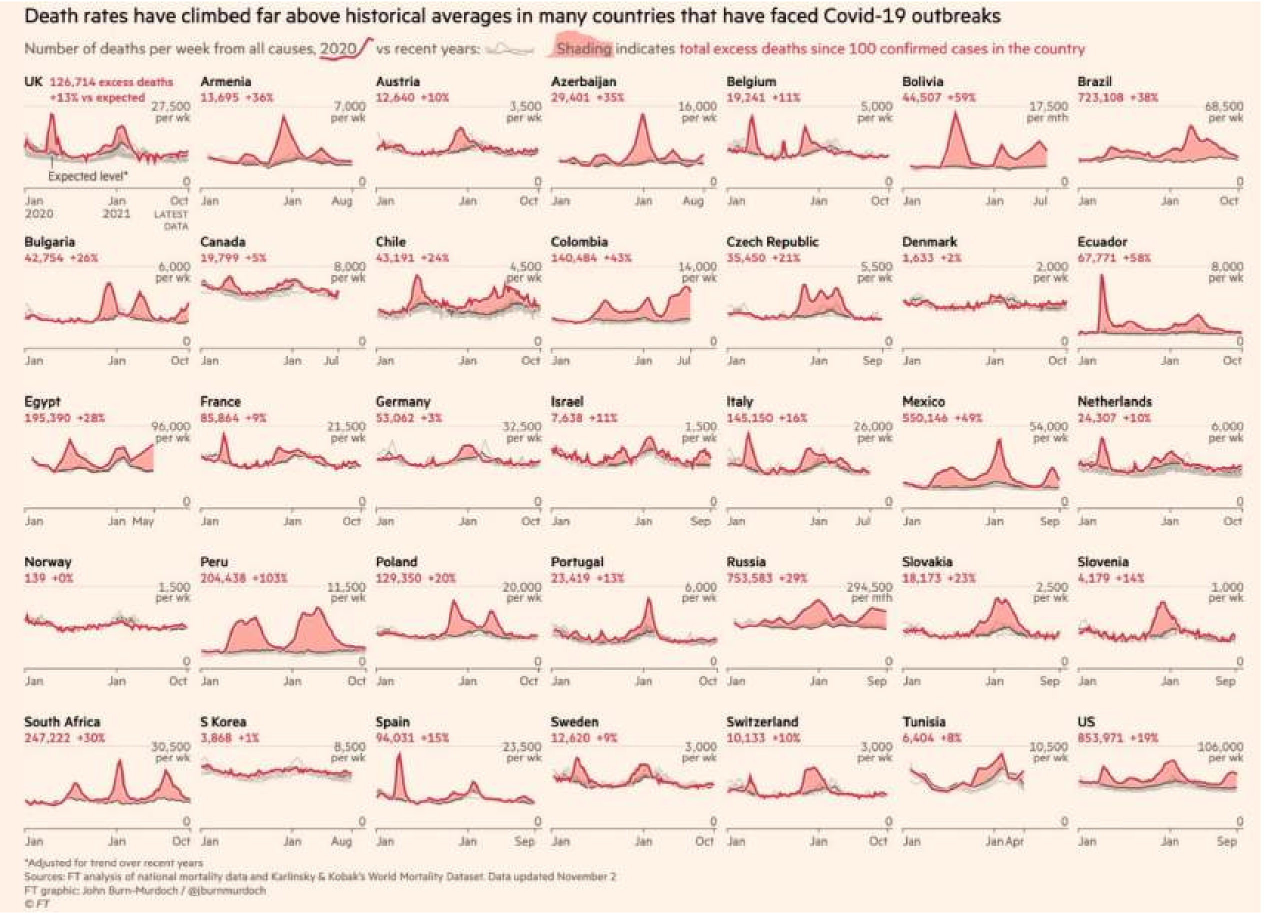


Source: David McCandless and Lee Bryon (2008)

Types of Graphical Displays

3. Small Multiples

Co-current series across categories.

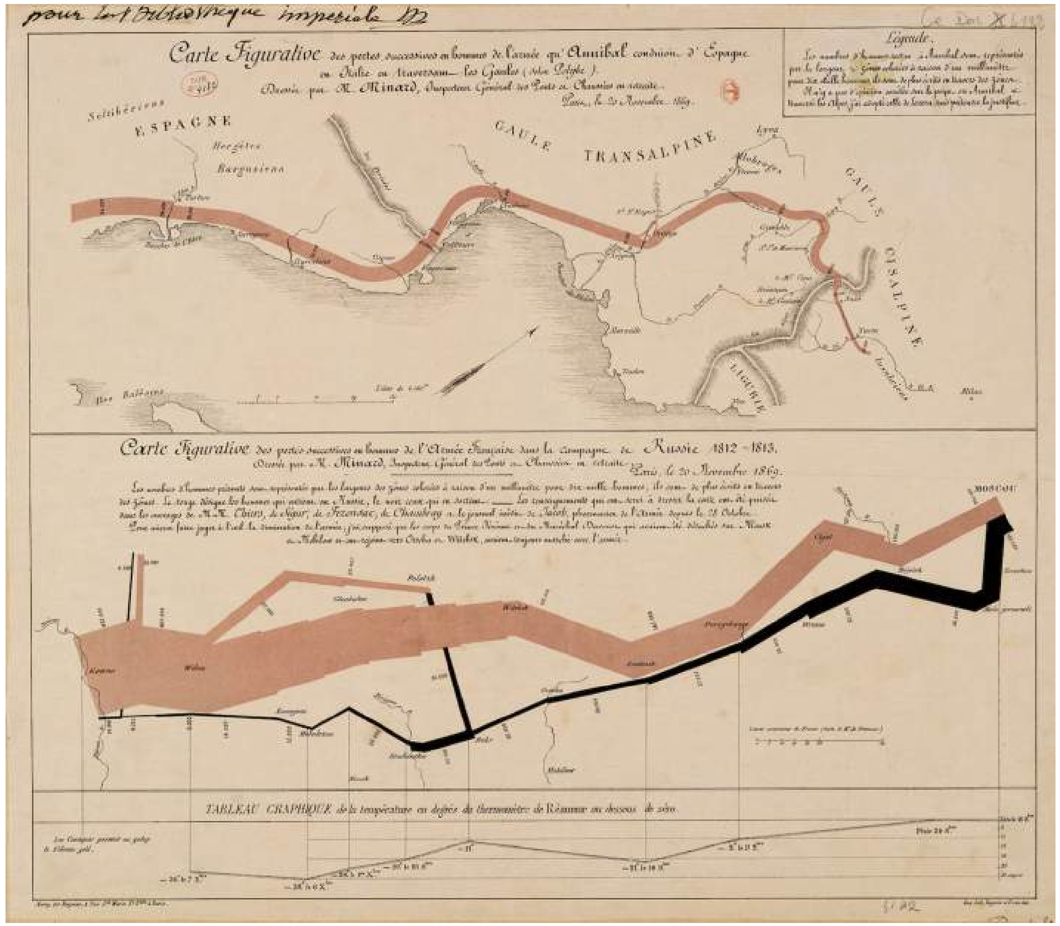


Source: Financial Times, 20 Dec 2021

Types of Graphical Displays

4. Narrative Graphics

Integrate multiple dimensions (space, time, quantity, and events) into a single coherent story.

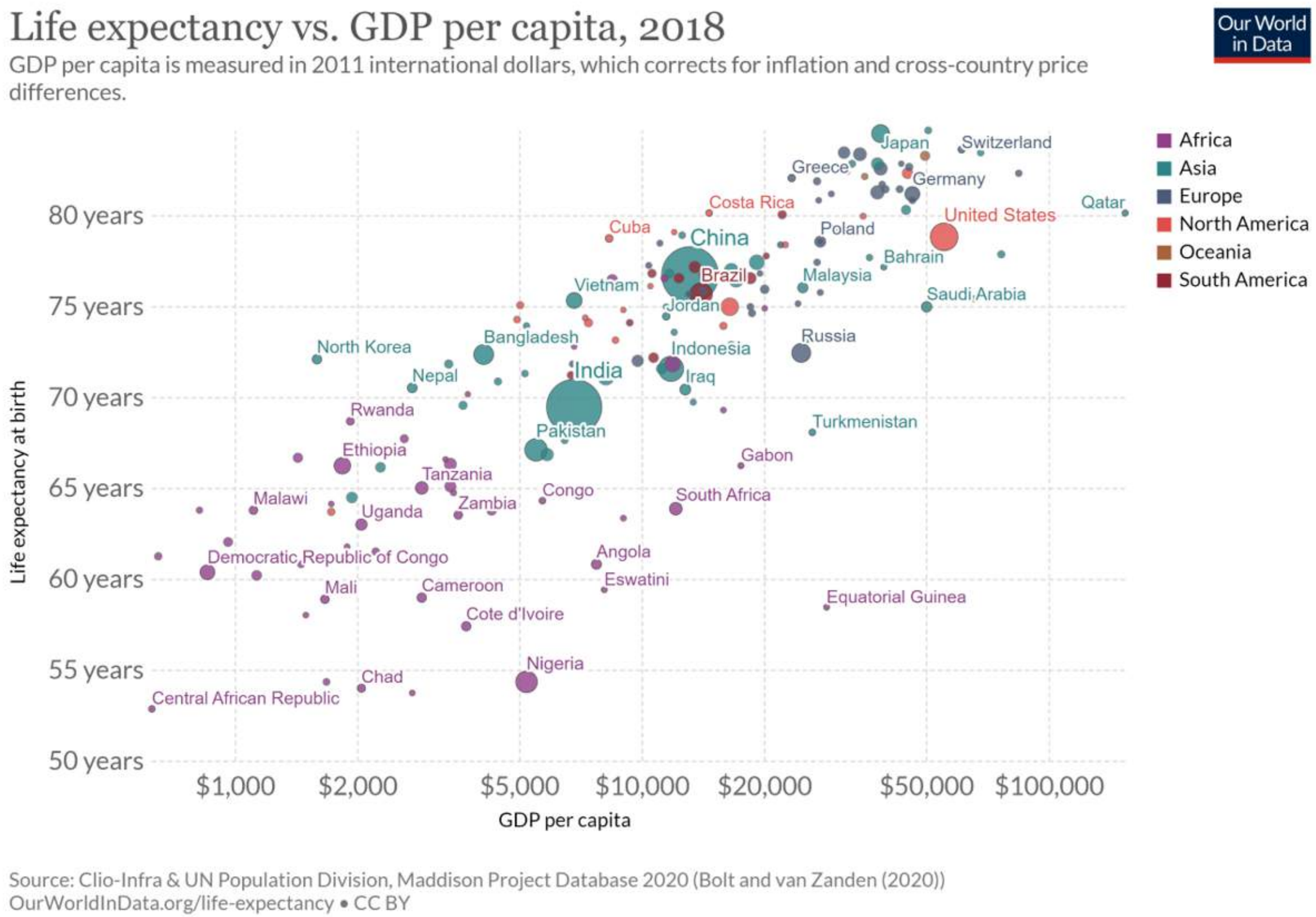


Source: Charles Minard (1869)

Types of Graphical Displays

5. Relational Data

Bivariate and multivariate relationships among variables.



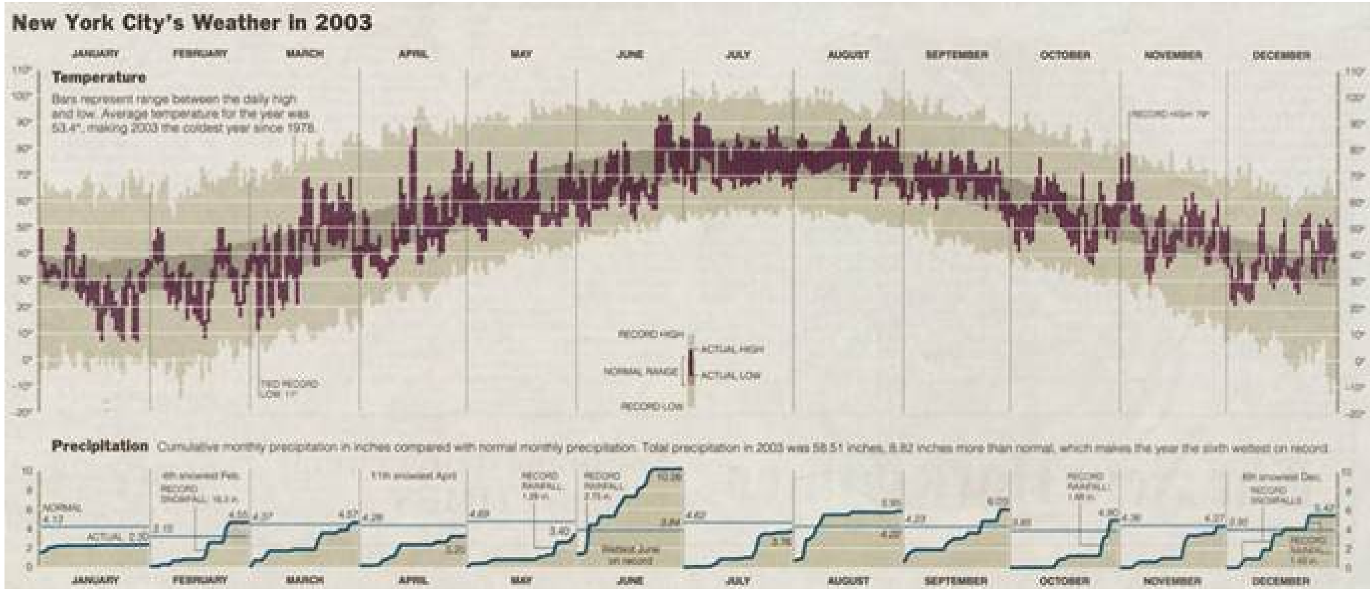
Source: Our World in Data

What Makes a Good Graph?

- **Show** the **data**.
- **Avoid distorting** what the data have to say.
- Present **many numbers** in a small space.
- Induce the viewer to **think** about the **substance** rather than the methodology, graphic design, or the software used, etc.
- Encourage comparison.
- Reveal data at **multiple levels**: from broad overview to fine-grained insights.
- Serve a clear **purpose**: Description, exploration, or tabulation.

Let’s take a look at some examples of good graphics.

Good Graphics: New York City Weather

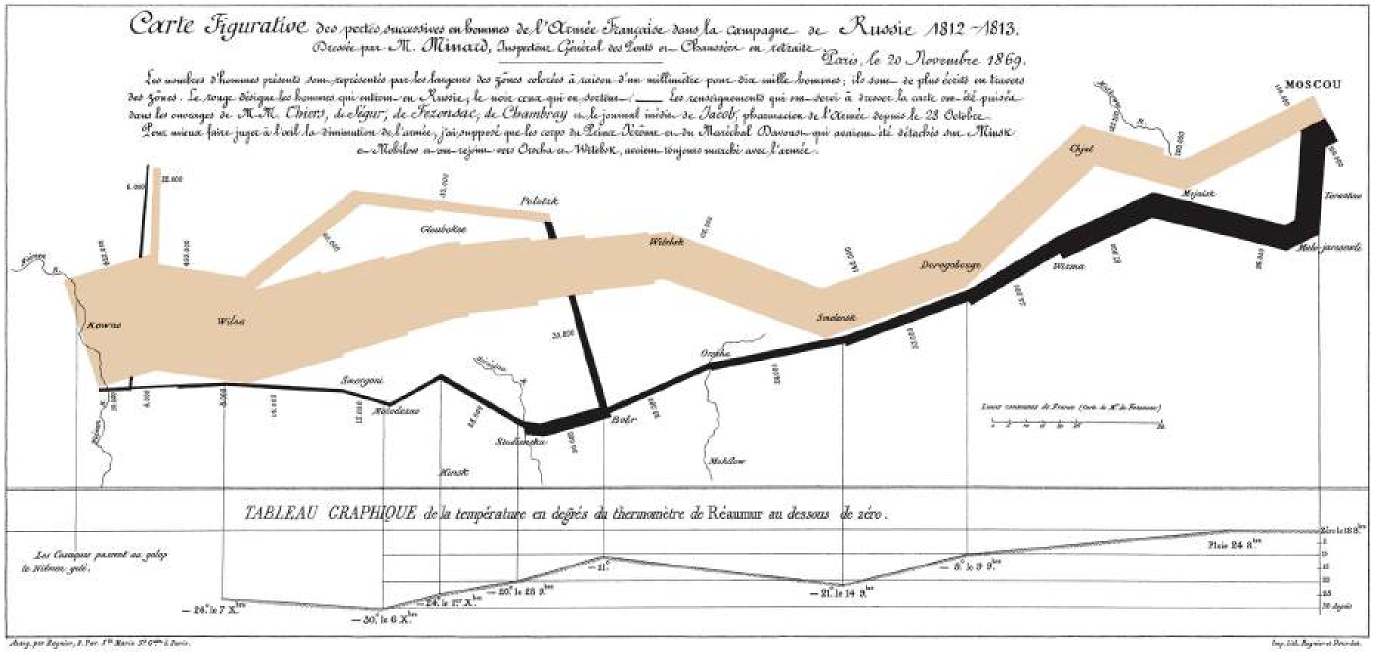


A time series graph that conveys different scales of information:

- How temperature varies over the year.
- How rainfall varies from month to month.
- Record extremes to provide historical context.

Source: Edward Tufte, New York Times Interactive

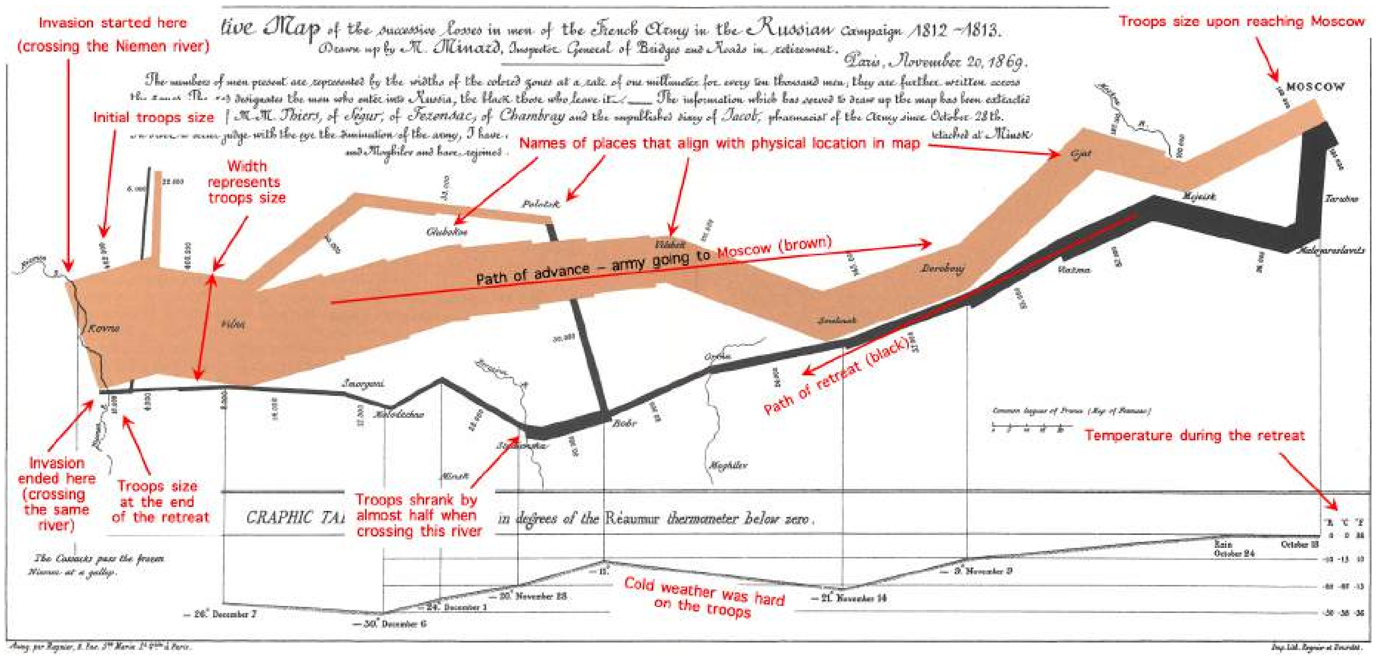
Good Graphics: Napoleon’s Army in Russia



- One of the most celebrated early graphs.
- The losses suffered by Napoleon’s army during the 1812 Russian Campaign.
- Multiple variables: Size of army, location, direction of movement, temperature, and various dates of during retreat.

Source: Edward Tufte

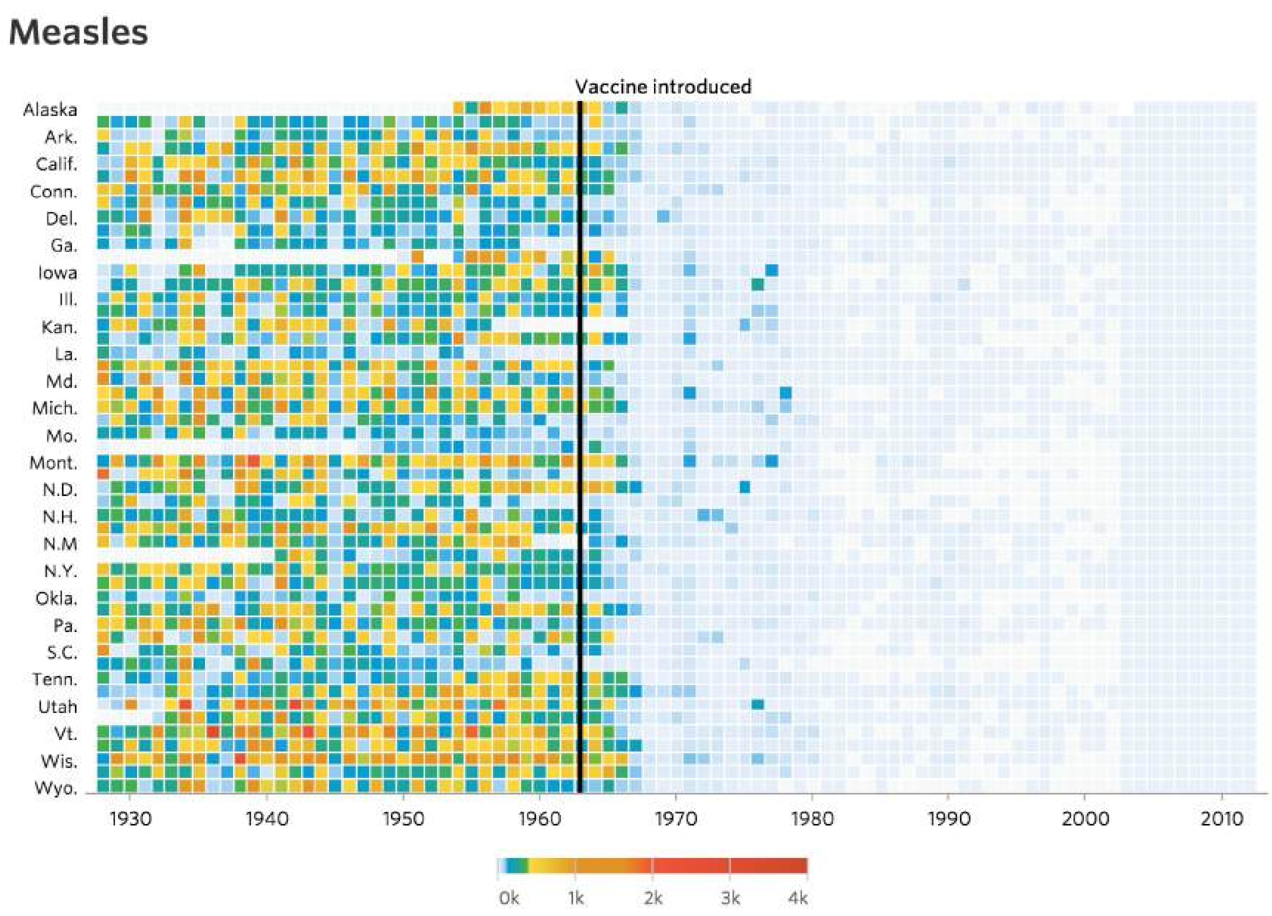
Good Graphics: Napoleon's Army in Russia



- In the beginning, he started with 422,000 soldiers.
- As the army advanced toward Moscow, the beige band narrowed sharply. Only 100,000 remained on arrival.
- On the way back, the numbers dropped even more. Fewer than 10,000 soldiers arrived home.

Good Graphics: Measles Infection

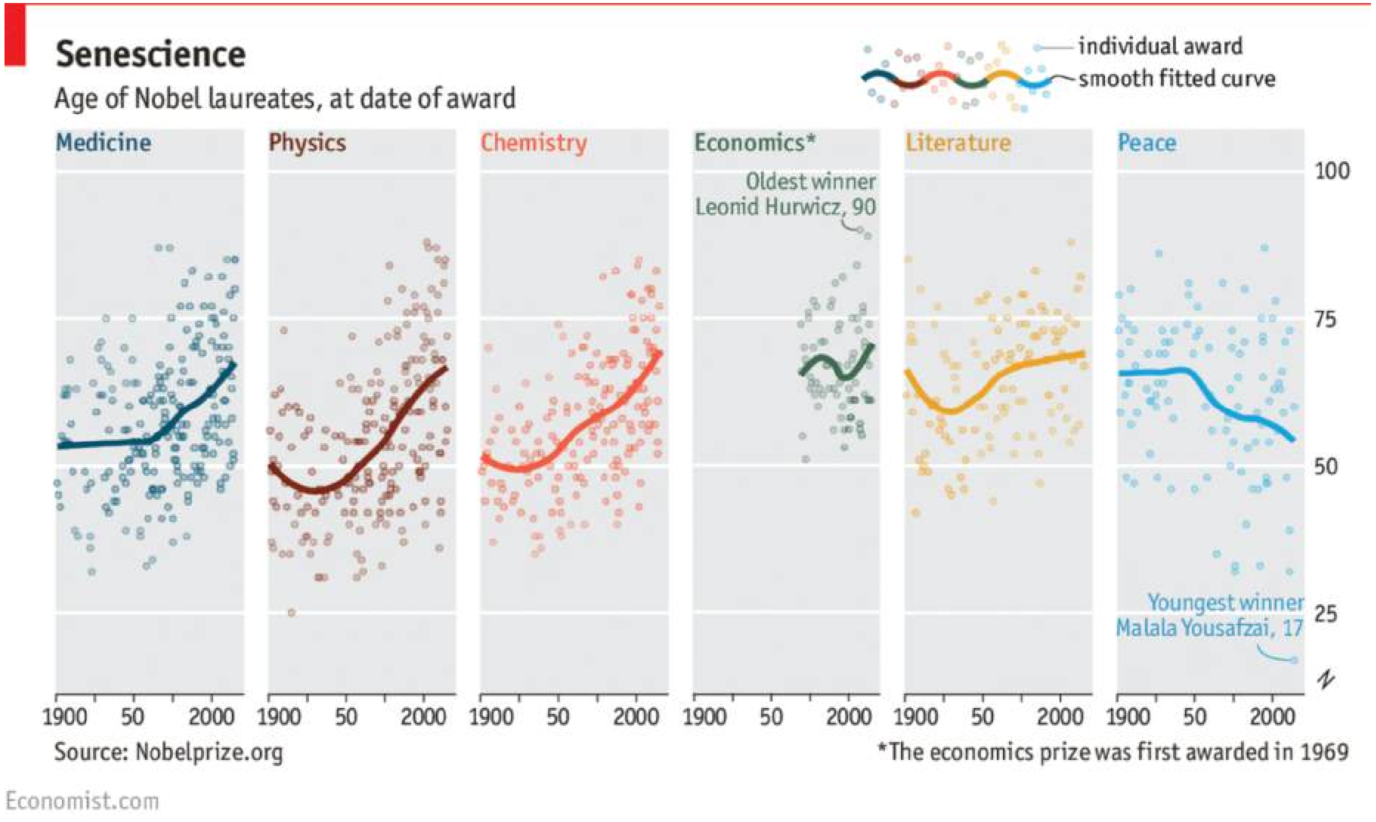
Notice the drop in infections after vaccines were introduced.



Source: Wall Street Journal, 11 Feb 2015

Good Graphics: Age of Nobel Laureates

The age of Nobel laureates by award category.



Source: The Economist, 3 Oct 2016

Graphical Excellence

- Well-designed presentation of interesting data.
- Communicate complex ideas with clarity, precision, and efficiency.
- Give the viewer the greatest number of ideas in the shortest time.
  - The best visuals are multivariate, showing relationships across multiple dimensions.
- Tell the truth about the data.

Bad Graphics

Let’s take a look at some bad graphics.

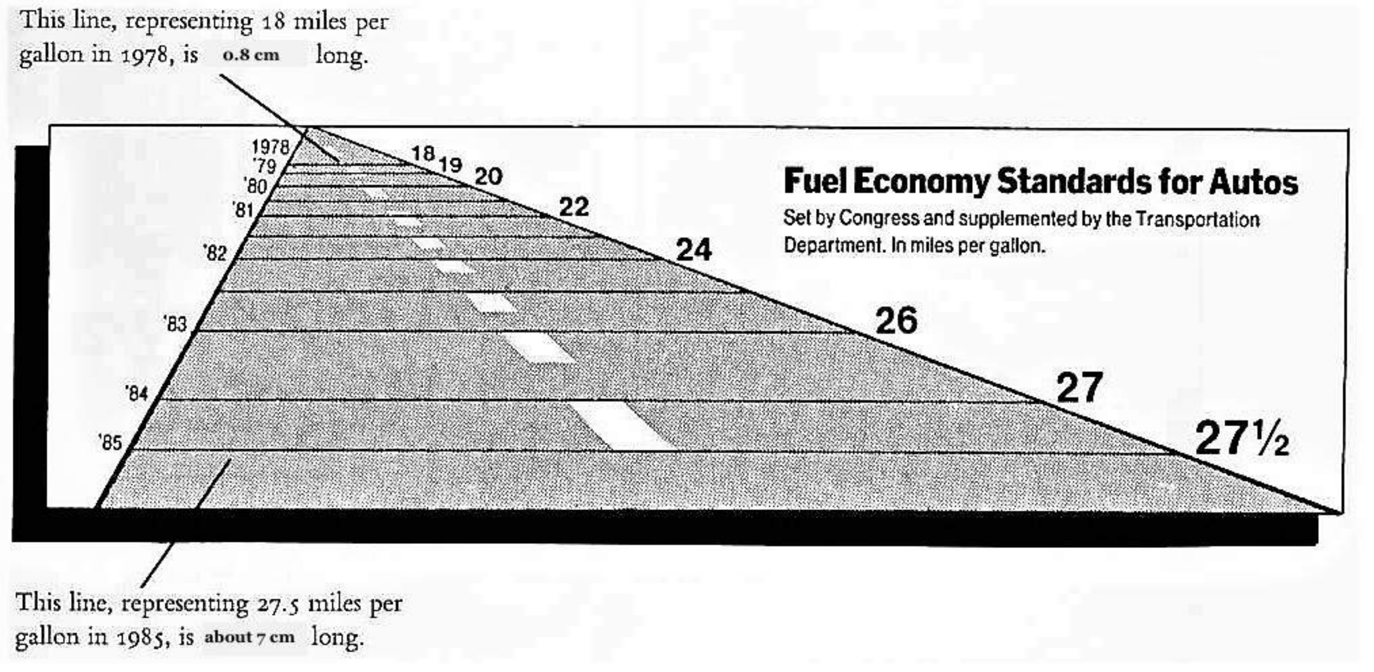
Problematic graphics tend to come in three varieties:

- Bad Perception
- Bad Data
- Bad Design (Bad Taste)

Bad Perception

Fuel Economy Standards

This graph meant to show improvements in fuel economy standards from 1978 to 1985. What's wrong with it?



Bad Perception

Fuel Economy Standards: Lie Factor

Numeric representation:

- 18 miles per gallon (mpg) in 1978.
- 27.5 mpg in 1985 (a 53% increase).

Visual representation:

- Line length changes from 0.8cm to 7cm, a 775% increase!
- The **lie factor** = Size of effect on graph / Size of effect in data.

$$\text{Lie Factor} = \frac{(7 - 0.8)/0.8}{(27.5 - 18)/18} \approx 14.7$$

Bad Perception

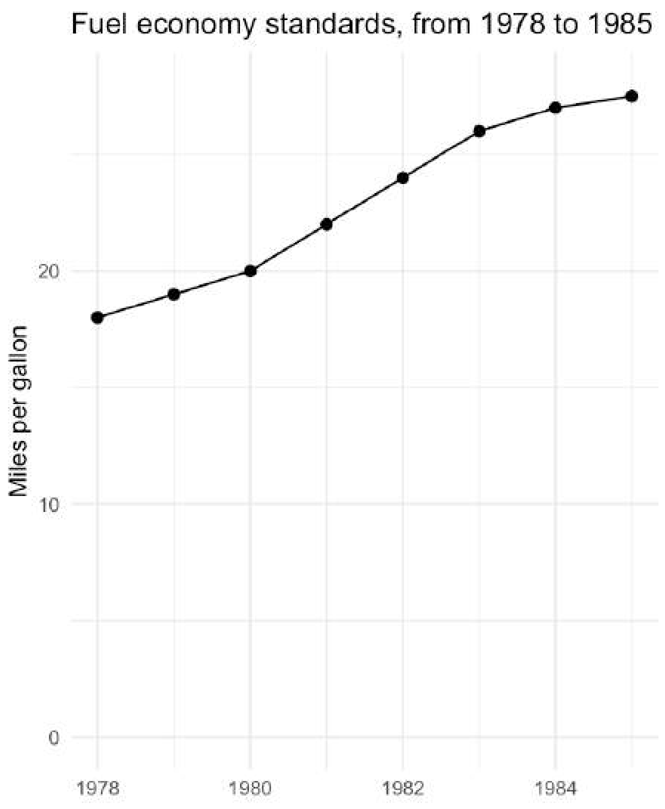
Fuel Economy Standards: Distortion

Each element of a graph creates visual expectations about its other parts.

- That is, 1cm on the axis on the left side of the graph should represent the same distance on the right side.
- If this consistency breaks, we have **distortion**.
- Distortion misleads the viewer, especially when visual differences appear more dramatic than the actual data.

Bad Perception

Fuel Economy Standards: An Honest Portrayal



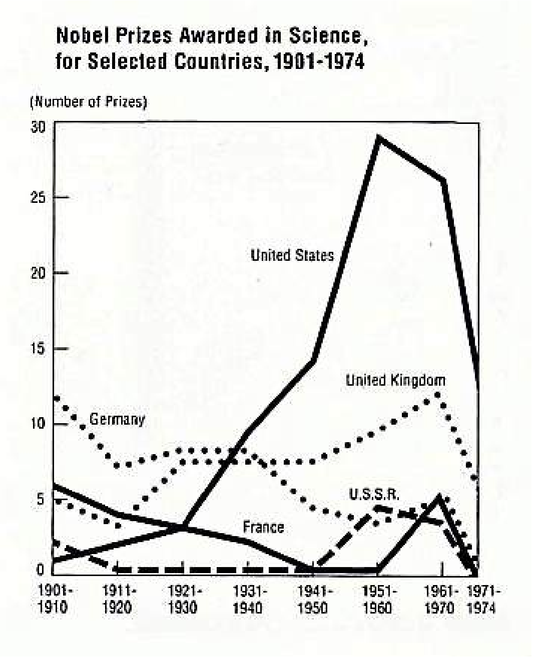
- The graph on the left is accurate.
- The increase in fuel economy standard is realistic and not exaggerated.
- It is our duty to ensure that the visual representation of the data is **consistent** with the numerical representation.

Bad Perception

Nobel Prizes: Inconsistent Basis of Comparison

A graph creates visual expectations – if the basis of comparison changes across time or across groups, the expectation can deceive the viewer.

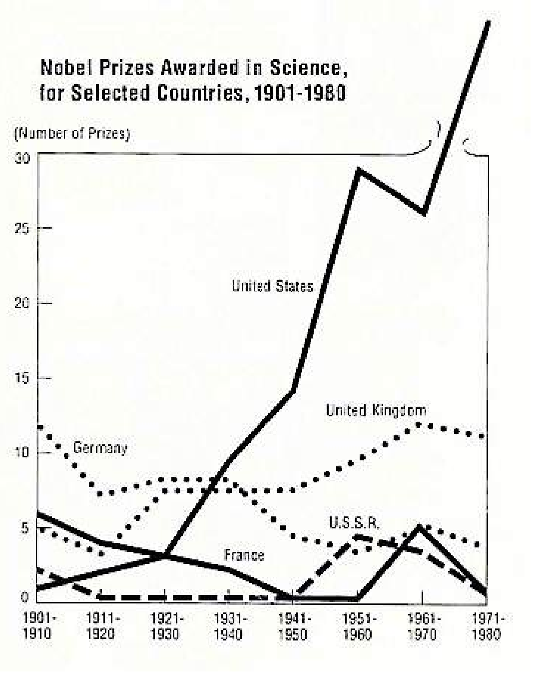
- The chart shows the number of Nobel Prizes awarded to selected countries.
- Notice the drastic decline in prizes by the United States in the most recent period.
- **But is it true?**



Bad Perception

Nobel Prizes: An Honest Portrayal

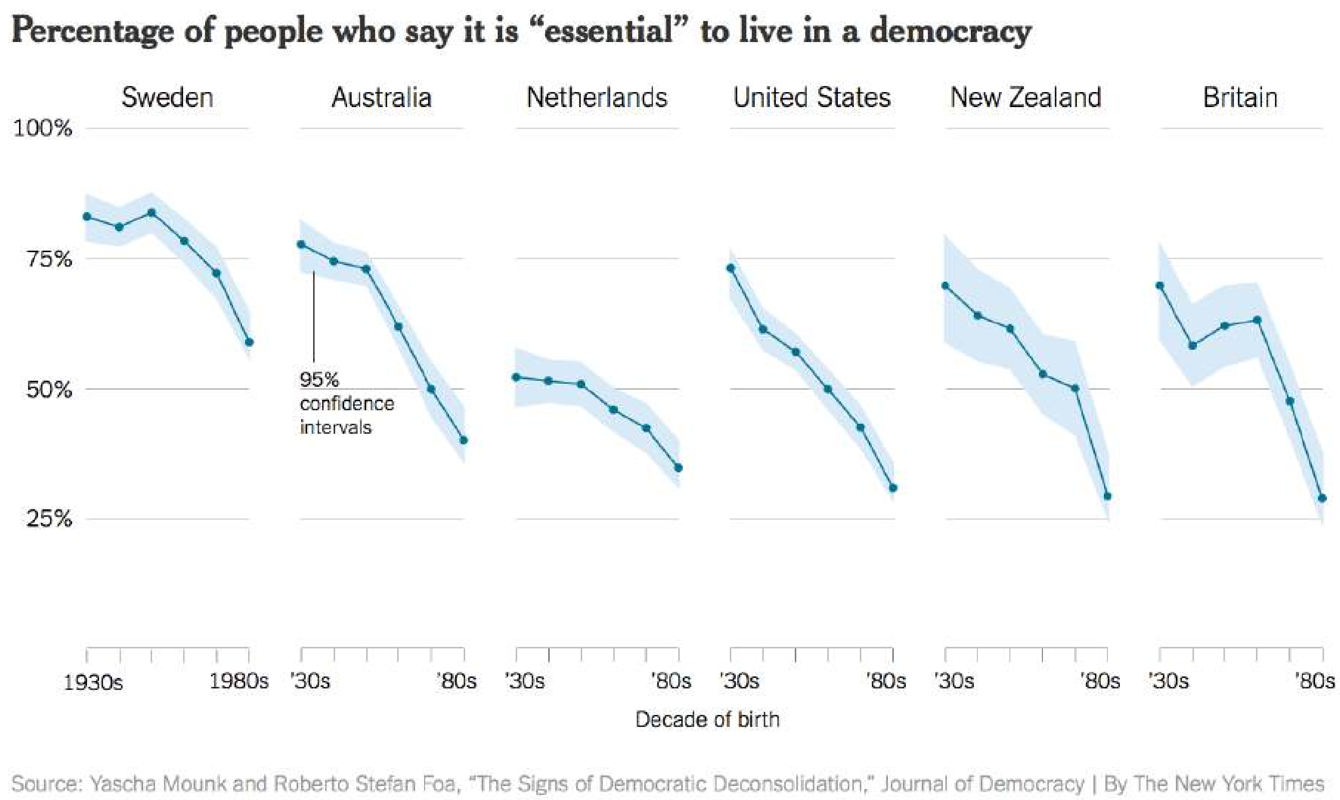
- **Look closer to the x-axis.**
- Each data point before 1970 represents a 10-year interval. But the final point covers only 1971-1974.
- Naturally, the count for this incomplete decade is lower.
- The graph's design gives the wrong impression of a sudden decline in Nobel Prizes.



Bad Data

Importance of Democracy

A well-designed figure can still mislead people if it builds on **bad data**.



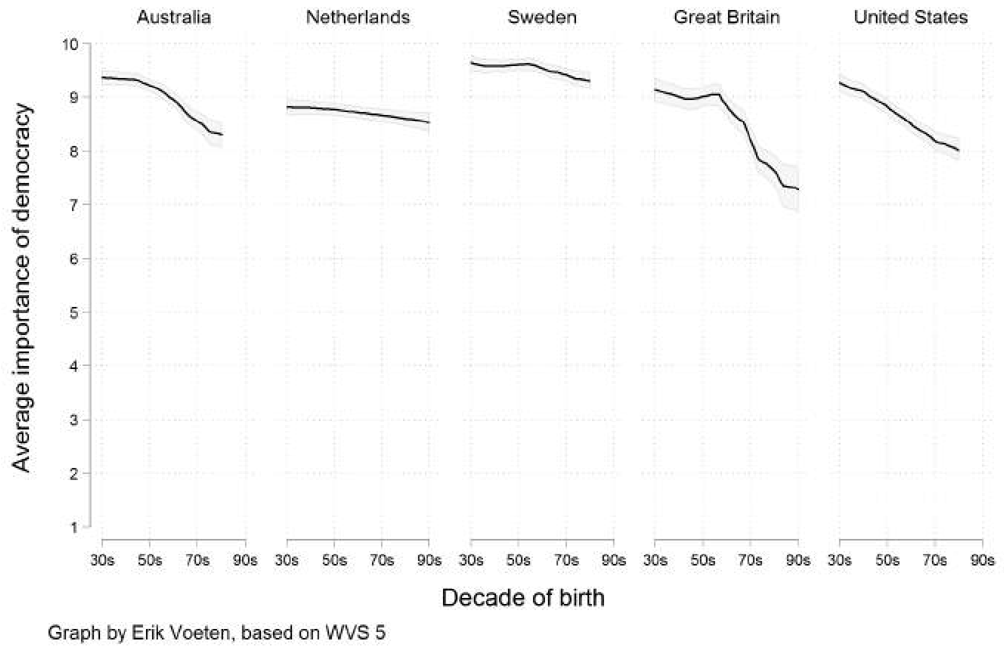
Source: New York Times, 29 Nov 2016

An alarming decline on people's confidence in democracy?

Bad Data

Importance of Democracy: Analysis

- Respondents asked to rate the importance of democracy on a 10-point scale.
- Original graph only showed the percentage of people who gave a score of 10.
- Here's an honest portrayal based on the average ratings:

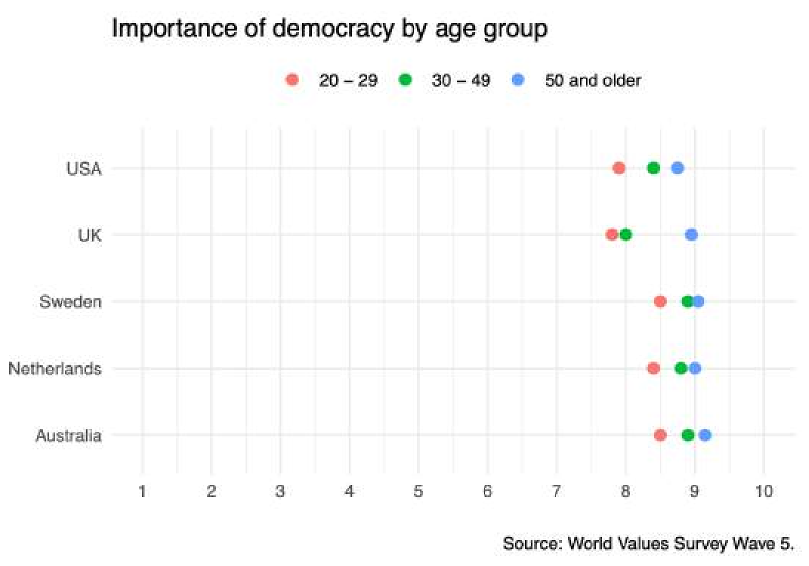


Source: Washington Post, 5 Dec 2016

Bad Data

Importance of Democracy: A Better Representation

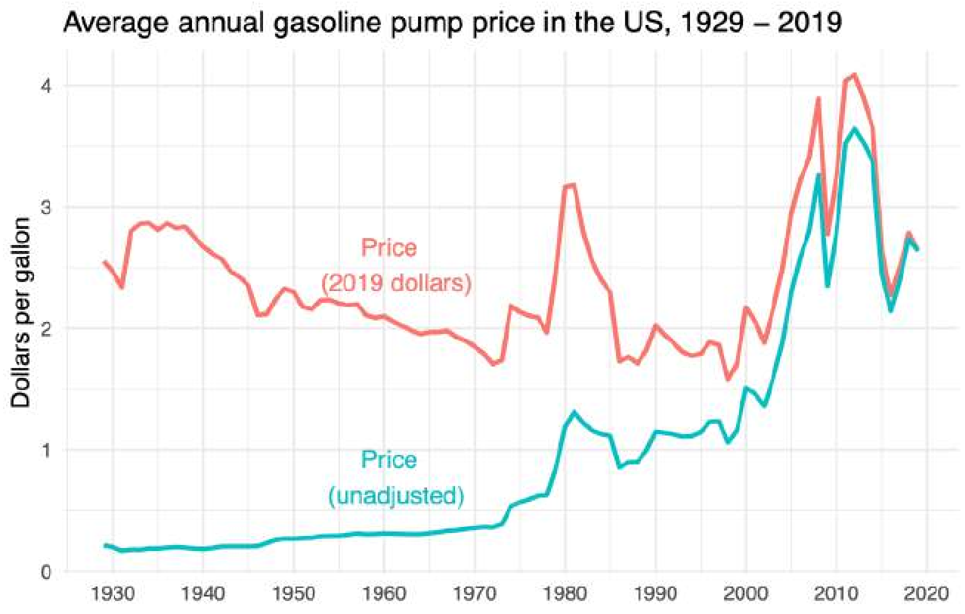
- The revised graph is still tricky to interpret.
- The x-axis does not represent time. It displays data from a cross-sectional survey of people of different ages.
- Not a trend measured at each decade from the 1930s.
- Rather, the differences reveal generational views, not changes over time.



Source: World Values Survey Wave 5

Bad Data

Time-Series Monetary Measurements



Source: U.S. Department of Energy

- When adjusted for inflation, the price of gasoline in 2019 was similar to the price in 1929.
- Failure to account for the high inflation would have depicted a worrying trend of increasing prices.

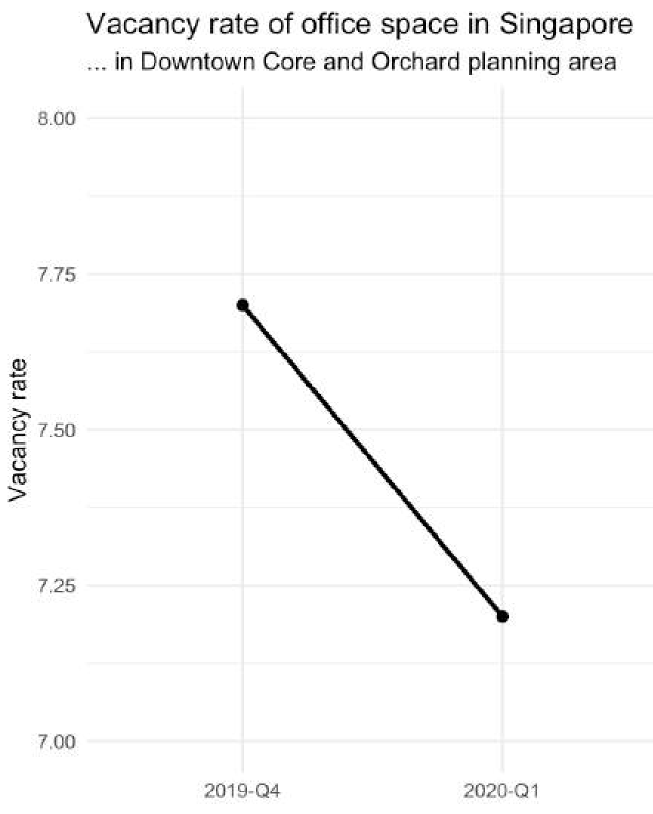
Bad Data

Context of Data

To be truthful and revealing, data graphics must bear on the question at the heart of quantitative thinking: “compared to what?”

Data-sparse graphics should provoke suspicion. Graphics often lie by omission.

Nearly all important questions are left unanswered by this display.

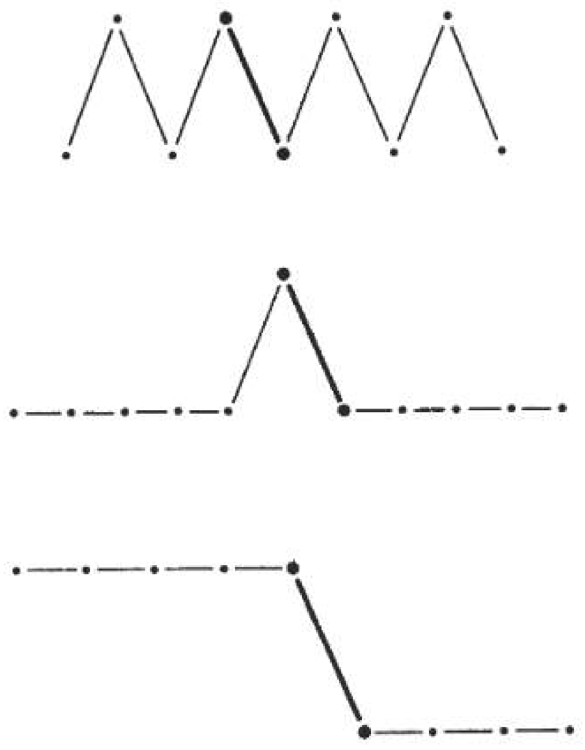


Source: Real Estate Information System (REALIS)

Bad Data

Context of Data: Possibilities

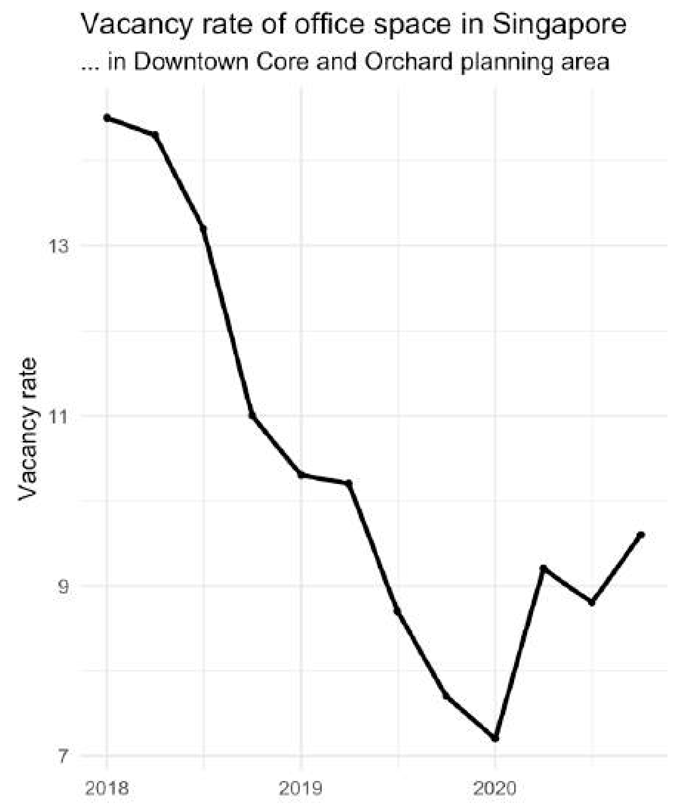
Imagine the very different interpretations other possible time-paths surrounding the 2019-20 change would have.



Bad Data

Context of Data: A Honest Portrayal

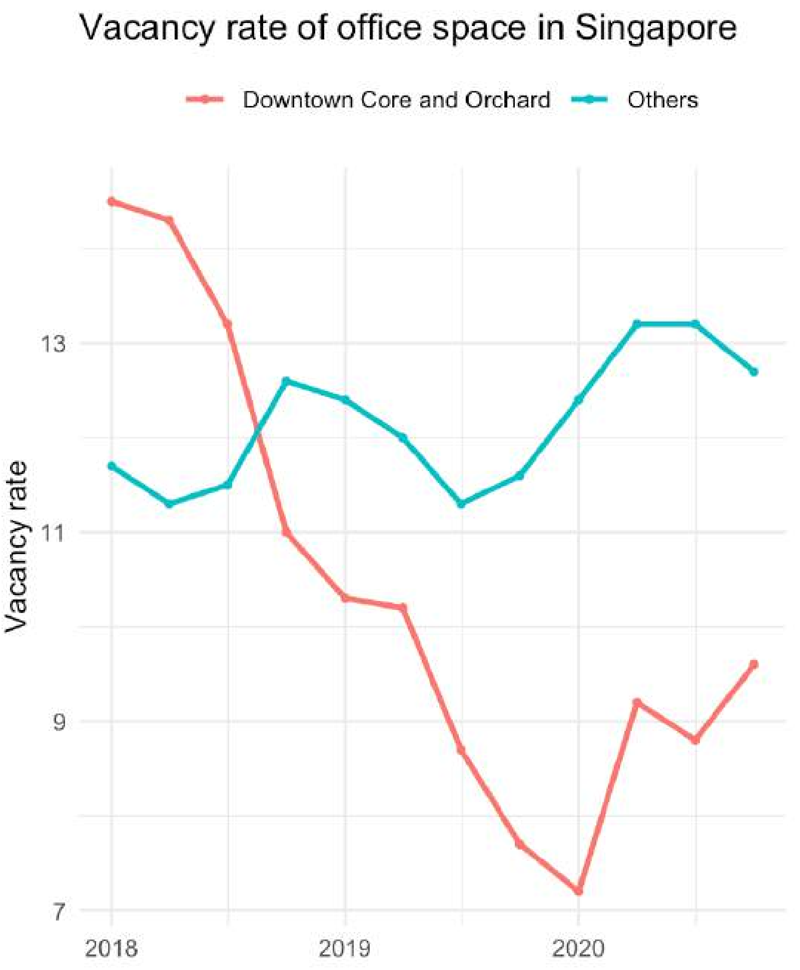
A few more data points add immensely to telling a more complete story.



Bad Data

Context of Data: A Honest Portrayal

Comparison with adjacent regions gives an even better context.



Bad Design

The Duck

Most of the time, interior decoration of graphics is unnecessary. Excessive visual elements are referred to as **chart junks**.

> Graphical decoration … comes cheaper than the hard work required to produce intriguing numebrs and secure evidence.

- When a graphic is taken over by decorative forms, it becomes a *duck*
  - The term comes from this duck-shaped building whose form overwhelms its function.
- It is okay to decorate construction, but **NEVER** construct decoration.

Source: Edward Tufte

Bad Design

Chart Junks

Source: Mike Cisneros

The most common types of **chart junk** are:

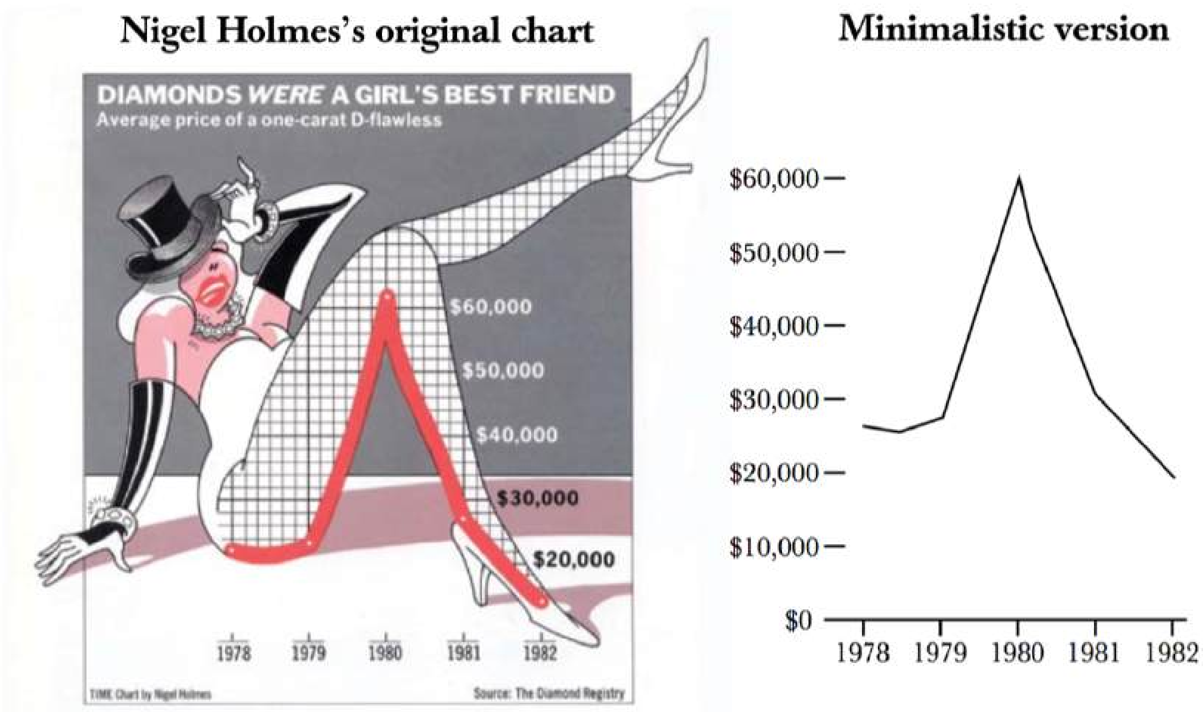
- Redundant elements that represent simple data.
- Unnecessary 3D and shadow effects.
- Overly-elaborated presentation.

Bad Design

Unnecessary Dimensions

Sometimes designers add visual elements that don’t serve the data.

- It creates distractions rather than insights.



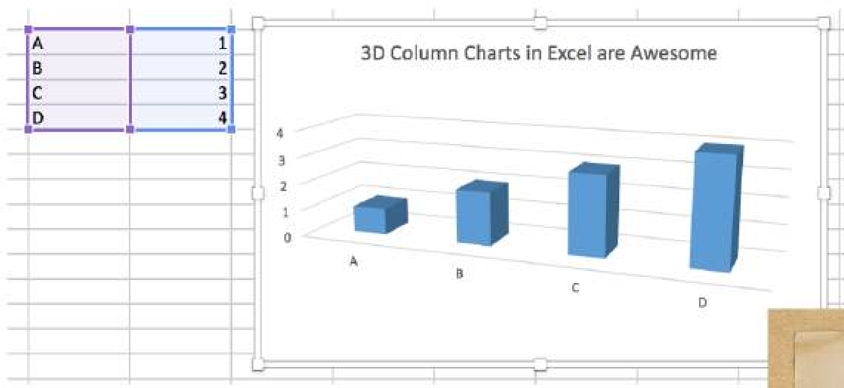
Source: Bateman et al. (2010)

Bad Design

Unnecessary 3D and Shadow Effects

3D effects distorts perception and hides meaning.

- Makes the plot difficult to read.
- It also adds complexity without extra information.

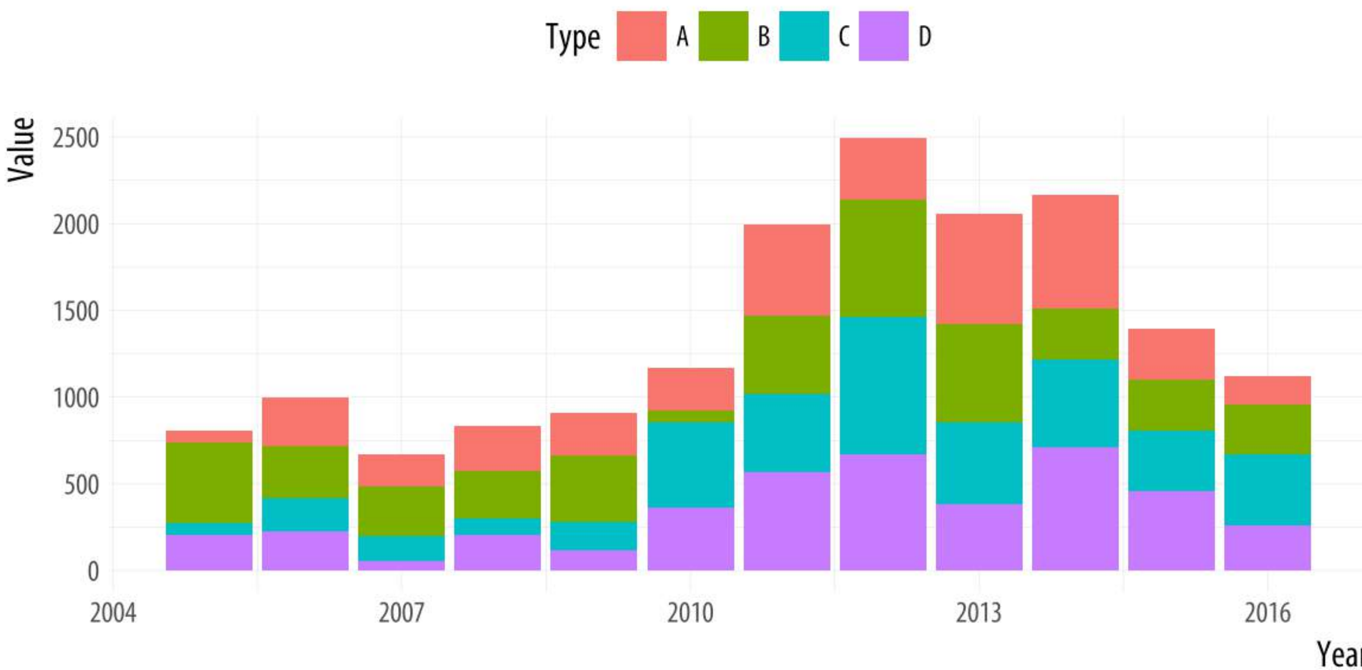


Bad Design

Overly-elaborated Presentation

A plot can be free of decoration, and still fail if its design makes it difficult to interpret.

- This stacked bar encodes multiple categories with colors.
- Difficult to compare values across years for a single category because the baseline is shifting.



Graphical Integrity

To maintain integrity and accuracy of visualisations:

1. The graphical representation of numbers should be directly proportional to the numerical quantities.
2. Clear, detailed, and thorough labeling should be used to defeat graphical distortion and ambiguity. If necessary, (a) use annotations to explain the data and (b) label important events.
3. Show data variation, not design variation.
4. Use standardised values.
    - In time-series data, prefer real (deflated) values over nominal values.
    - For comparisons, standardise scales (e.g., per capta, per 100,000 people).
5. The number of information-carrying dimensions depicted should not exceed the number of dimensions in the data.
6. Graphics must quote data in context.

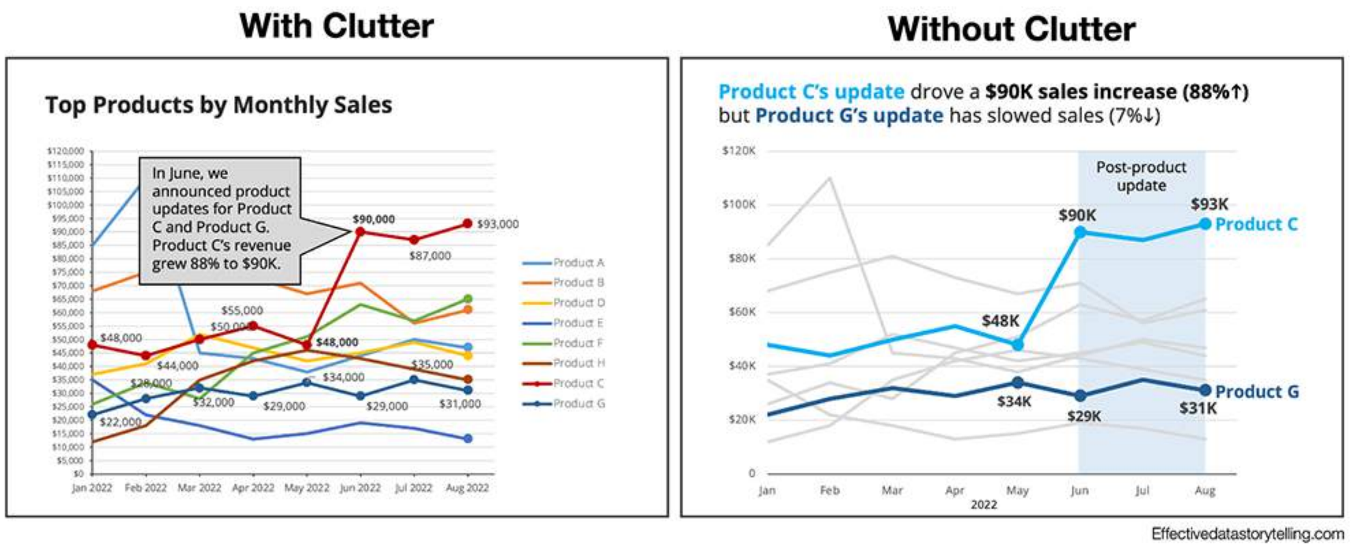
Graphical Excellence

Data-ink Ratio

$$\text{Data-ink Ratio} = \frac{\text{data-ink}}{\text{total ink used to print the graph}}$$

The goal is to **maximise data ink**,

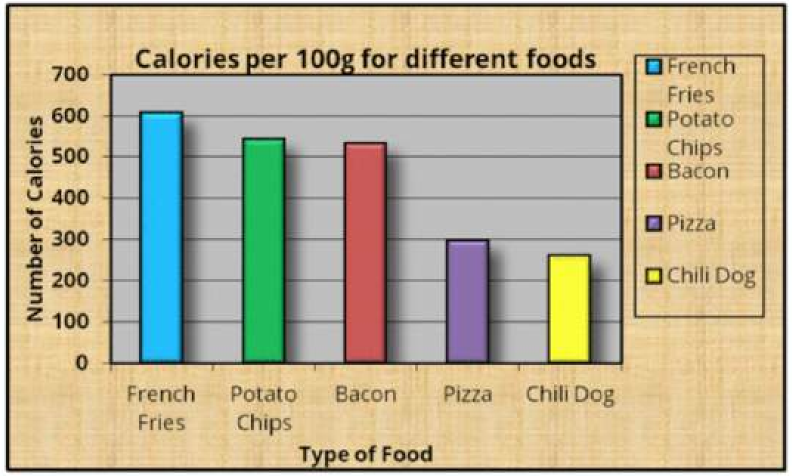
- … the portion of visual elements that represent actual information, not decoration.

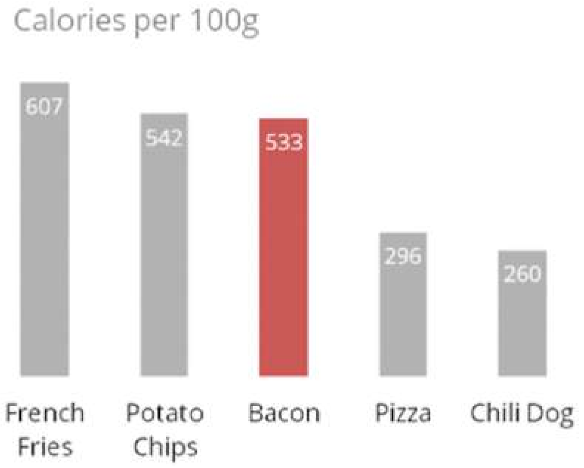


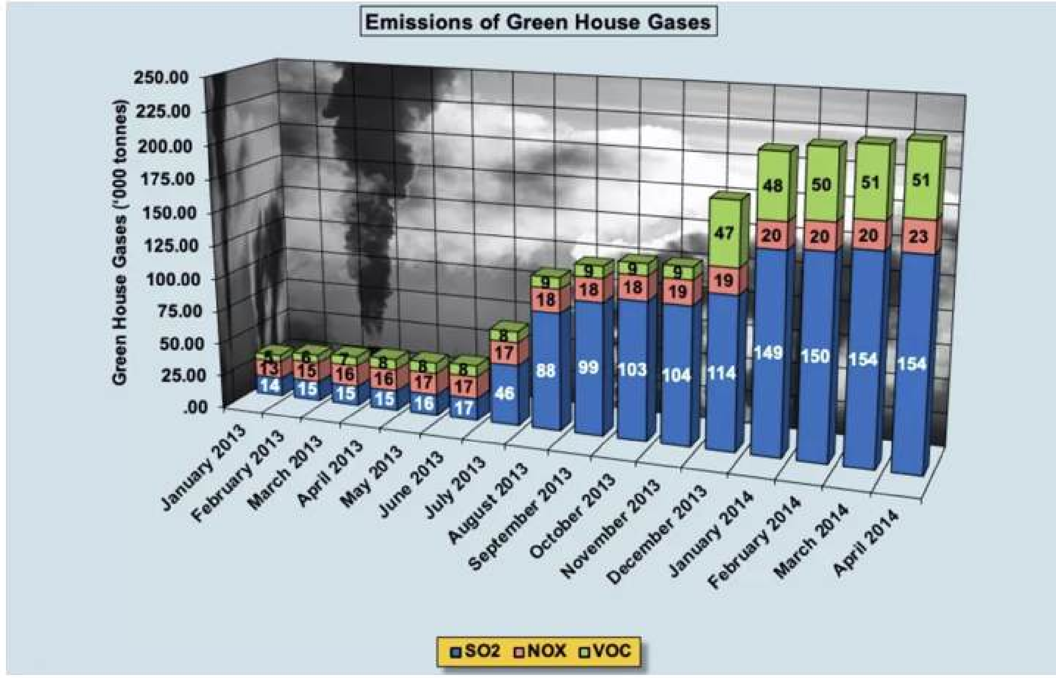
Data-ink Ratio

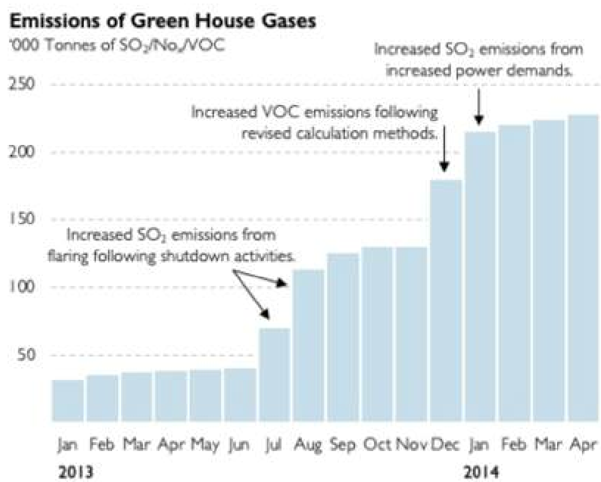
Humans are poor at judging depth and angle.

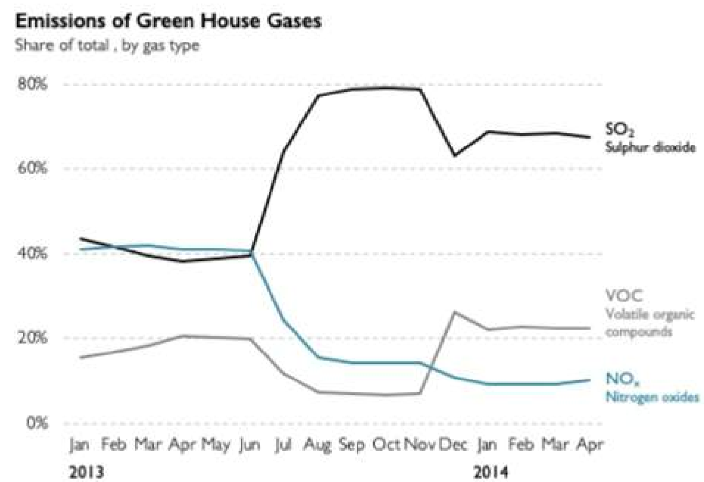
- 3D charts often distort proportions.
- It would be better use flat 2D bars or lines to represent the values, with color or shades to distinguish categories.






Emissions of Green House Gases
Green House Gases ('000 tonnes)
250.00
225.00
200.00
175.00
150.00
125.00
100.00
75.00
50.00
25.00
.00
January 2013
February 2013
March 2013
April 2013
May 2013
June 2013
July 2013
August 2013
September 2013
October 2013
November 2013
December 2013
January 2014
February 2014
March 2014
April 2014
SO2
NOX
VOC

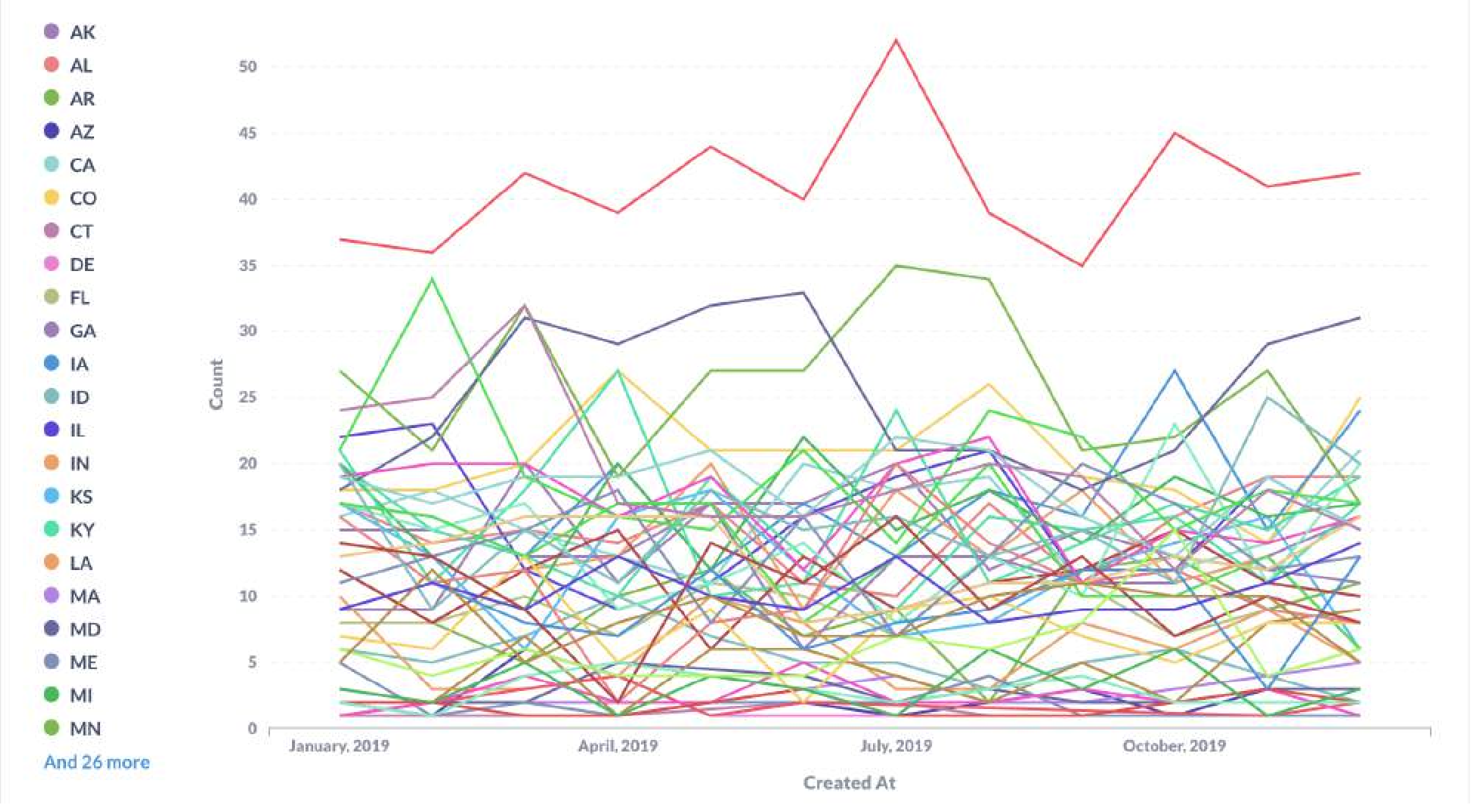

Emissions of Green House Gases
'000 Tonnes of SO_2/No_x/VOC
Increased SO_2 emissions from increased power demands.
Increased VOC emissions following revised calculation methods.
Increased SO_2 emissions from flaring following shutdown activities.
250
200
150
100
50
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr
2013
2014


Emissions of Green House Gases
Share of total , by gas type
80%
60%
40%
20%
0%
SO_2
Sulphur dioxide
VOC
Volatile organic compounds
NO_x
Nitrogen oxides
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr
2013
2014

Too Many ~~Cooks~~ Lines

Do we really need a separate line (and color) for each of the 50 states?

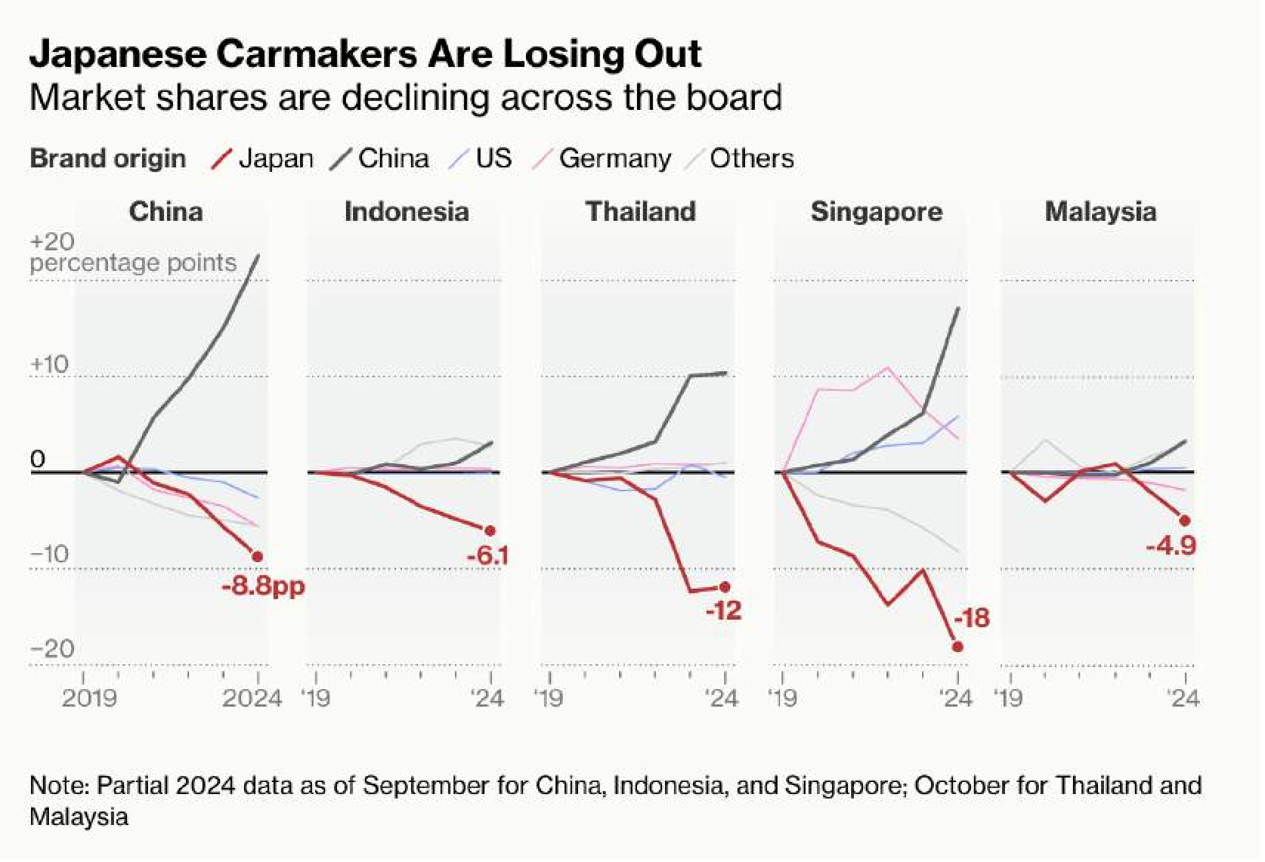
- Too many series plotted together result in *visual noise*, not insight.
- Over-encoding increase total ink to print the graph, without increasing information.



Better Approach: Small Multiples

Multiple time series can be effective - improve clarity by highlighting one key group at a time.

One color draw focus, others provide context.

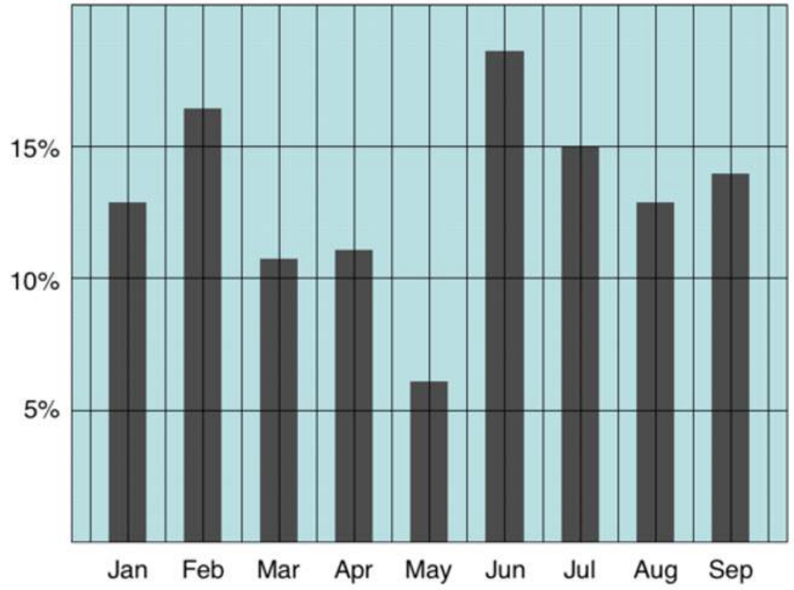


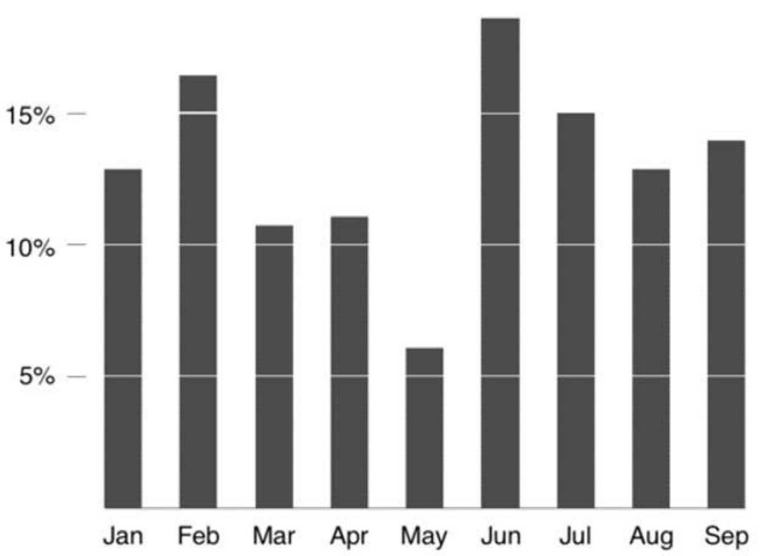
Source: Bloomberg (2024)

Reference Lines

... improves readability by allowing viewers to estimate values easily.

But they should be subtle and unobtrusive.





Tables

Tables should be readable but visually clean.

- Grid lines aid alignment, but excessive lines make the table difficult to read.
- In most situations, keep subtle horizontal lines and remove vertical ones.

Short-term credit lines by bank

June 30, 1980 (millions USD)

	Finance Companies					Manufacturing Companies							
Bank	Canada	Germany	UK	USA	*Total*	Argentina	Australia	Canada	France	UK	USA	*Total*	*Grand Total*
Allied & Associates	0.0	0.0	18.6	0.0	*18.6*	0.0	0.0	0.0	0.0	0.0	0.0	*0.0*	*18.6*
Bank of America	0.0	0.0	0.0	17.5	*17.5*	0.0	3.3	0.0	0.0	0.0	17.5	*20.8*	*38.3*
Bankers Trust	0.0	0.0	0.0	13.0	*13.0*	0.0	0.0	0.0	0.0	0.0	13.0	*13.0*	*26.0*
Banque National de Paris	0.0	0.0	11.3	0.0	*11.3*	0.0	0.0	15.0	34.6	0.0	0.0	*49.6*	*60.9*
Barclays	0.0	0.0	42.5	0.0	*42.5*	0.0	0.0	0.0	0.0	133.4	0.0	*133.4*	*175.9*
Chase Manhattan	0.0	0.0	0.0	15.2	*15.2*	0.0	0.0	0.0	0.0	0.0	15.0	*15.0*	*30.2*
Chemical	0.0	0.0	0.0	13.3	*13.3*	0.0	0.0	0.0	0.0	0.0	13.0	*13.0*	*26.3*
CIBC	37.8	0.0	27.1	0.0	*64.9*	0.0	0.0	222.9	3.6	3.4	0.0	*229.9*	*294.8*
Citibank	0.0	0.0	0.0	17.5	*17.5*	0.5	0.0	0.0	0.0	1.1	17.5	*19.1*	*36.6*
Commerzbank	0.0	3.3	0.0	0.0	*3.3*	0.0	0.0	0.0	0.0	0.0	0.0	*0.0*	*3.3*
Continental Illinois	0.0	0.0	4.6	21.8	*26.4*	0.0	0.0	0.0	0.0	0.0	21.1	*21.1*	*47.5*
Crédit Lyonnais	0.0	0.0	9.0	0.0	*9.0*	0.0	0.0	0.0	33.0	0.0	3.0	*36.0*	*45.0*
Deutsche Bank	0.0	3.3	0.0	0.0	*3.3*	0.0	0.0	0.0	0.0	0.0	0.0	*0.0*	*3.3*
Dresdner	0.0	3.3	0.0	0.0	*3.3*	0.0	0.0	0.0	0.0	0.0	0.0	*0.0*	*3.3*
FNB Chicago	0.0	0.0	0.0	15.0	*15.0*	0.0	0.0	0.0	0.0	0.0	15.0	*15.0*	*30.0*
Lloyds	0.0	0.0	42.5	0.0	*42.5*	0.0	0.0	0.0	0.0	36.2	0.0	*36.2*	*78.7*
Midland	0.0	0.0	54.3	0.0	*54.3*	0.0	0.0	0.0	0.0	36.2	0.0	*36.2*	*90.5*
Royal Bank of Canada	0.0	0.0	11.3	0.0	*11.3*	0.0	0.0	0.0	0.0	0.0	15.0	*15.0*	*26.3*
Société General	0.0	0.0	9.0	0.0	*9.0*	0.0	0.0	0.0	50.1	0.0	0.0	*50.1*	*59.1*
Toronto Dominion	0.0	0.0	6.8	0.0	*6.8*	0.0	0.0	0.0	0.0	0.0	15.2	*15.2*	*22.0*
Others	0.0	7.8	50.8	78.1	*136.7*	40.1	25.5	0.0	16.1	19.2	48.2	*149.1*	*285.8*
Total	***37.8***	***17.7***	***287.8***	***191.4***	*534.7*	***40.6***	***28.8***	***237.9***	***137.4***	***229.5***	***193.5***	*867.7*	***1,402.4***

Short-term credit lines by bank

June 30, 1980 (millions USD)

		MANUFACTURING COMPANIES							FINANCE COMPANIES				
Bank	Grand Total	Canada	UK	USA	France	Argentina	Australia	Total	UK	USA	Canada	Germany	Total
Total	**1,402.4**	**237.9**	**229.5**	**193.5**	**137.4**	**40.6**	**28.8**	**867.7**	**287.8**	**191.4**	**37.8**	**17.7**	**534.7**
CIBC	294.8	222.9	3.4	-	3.6	-	-	229.9	27.1	-	37.8	-	64.9
Others	285.8	-	19.2	48.2	16.1	40.1	25.5	149.1	50.8	78.1	-	7.8	136.7
Barclays	175.9	-	133.4	-	-	-	-	133.4	42.5	-	-	-	42.5
Midland	90.5	-	36.2	-	-	-	-	36.2	54.3	-	-	-	54.3
Lloyds	78.7	-	36.2	-	-	-	-	36.2	42.5	-	-	-	42.5
Banque National de Paris	60.9	15.0	-	-	34.6	-	-	49.6	11.3	-	-	-	11.3
Société General	59.1	-	-	-	50.1	-	-	50.1	9.0	-	-	-	9.0
Continental Illinois	47.5	-	-	21.1	-	-	-	21.1	4.6	21.8	-	-	26.4
Crédit Lyonnais	45.0	-	-	3.0	33.0	-	-	36.0	9.0	-	-	-	9.0
Bank of America	38.3	-	-	17.5	-	-	3.3	20.8	-	17.5	-	-	17.5
Citibank	36.6	-	1.1	17.5	-	0.5	-	19.1	-	17.5	-	-	17.5
Chase Manhattan	30.2	-	-	15.0	-	-	-	15.0	-	15.2	-	-	15.2
FNB Chicago	30.0	-	-	15.0	-	-	-	15.0	-	15.0	-	-	15.0
Royal Bank of Canada	26.3	-	-	15.0	-	-	-	15.0	11.3	-	-	-	11.3
Chemical	26.3	-	-	13.0	-	-	-	13.0	-	13.3	-	-	13.3
Bankers Trust	26.0	-	-	13.0	-	-	-	13.0	-	13.0	-	-	13.0
Toronto Dominion	22.0	-	-	15.2	-	-	-	15.2	6.8	-	-	-	6.8
Allied & Associates	18.6	-	-	-	-	-	-	-	18.6	-	-	-	18.6
Commerzbank	3.3	-	-	-	-	-	-	-	-	-	-	3.3	3.3
Deutsche Bank	3.3	-	-	-	-	-	-	-	-	-	-	3.3	3.3
Dresdner	3.3	-	-	-	-	-	-	-	-	-	-	3.3	3.3

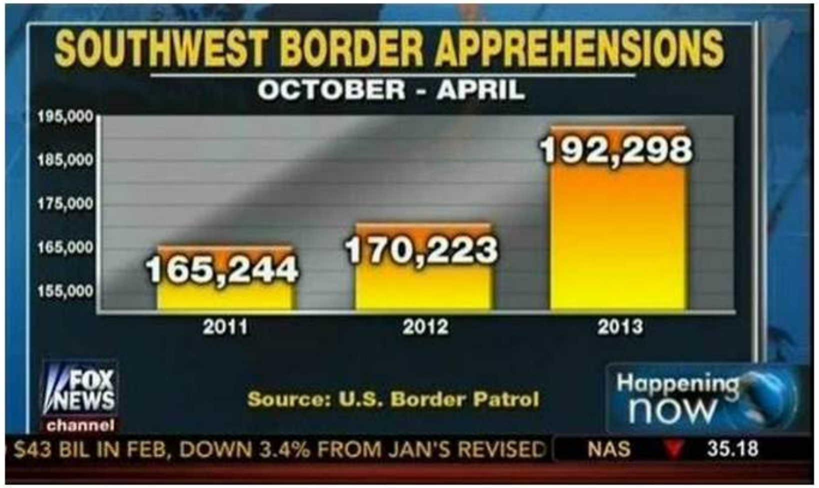
Graphical Excellence: Other Considerations

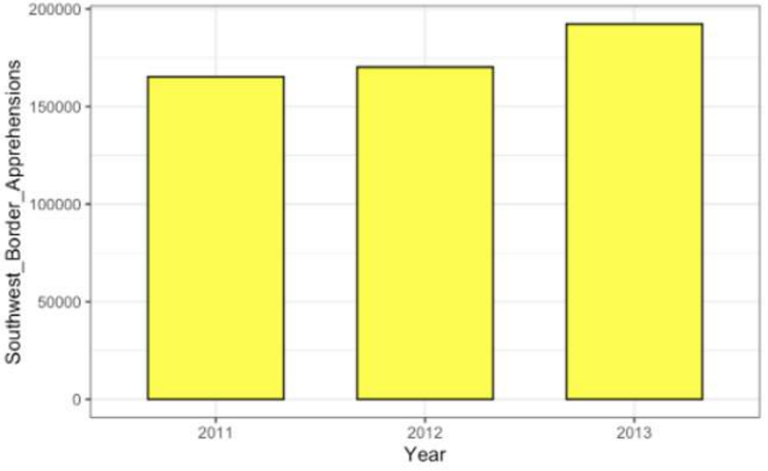
- Include the baseline.
- Avoid using area/angle/diameter to represent data.
- Data density.
- Consider your audience.
- Explain encodings.
- Avoid cherry-picking.
- Pre-attentive attributes and color theory

Include the Baseline

When using bar plots, we should start the bars at zero.

- Truncated y-axis will mislead the viewers into thinking that patterns are larger (or smaller) than they really are.



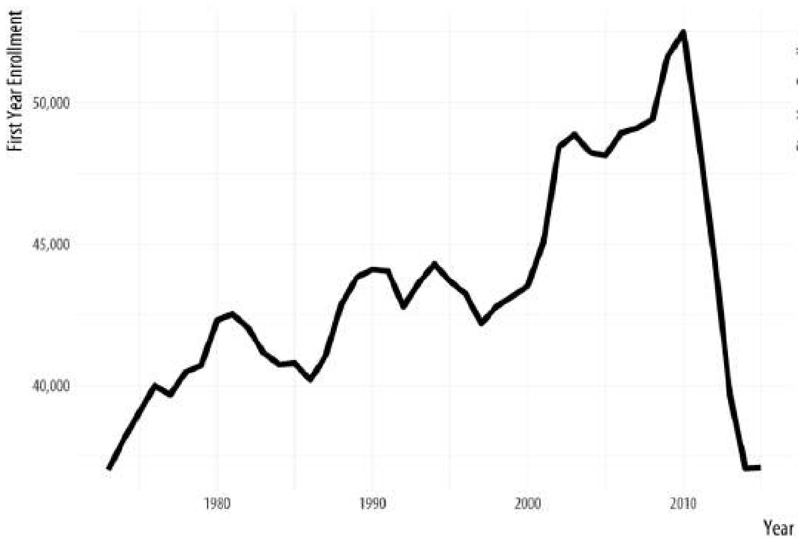


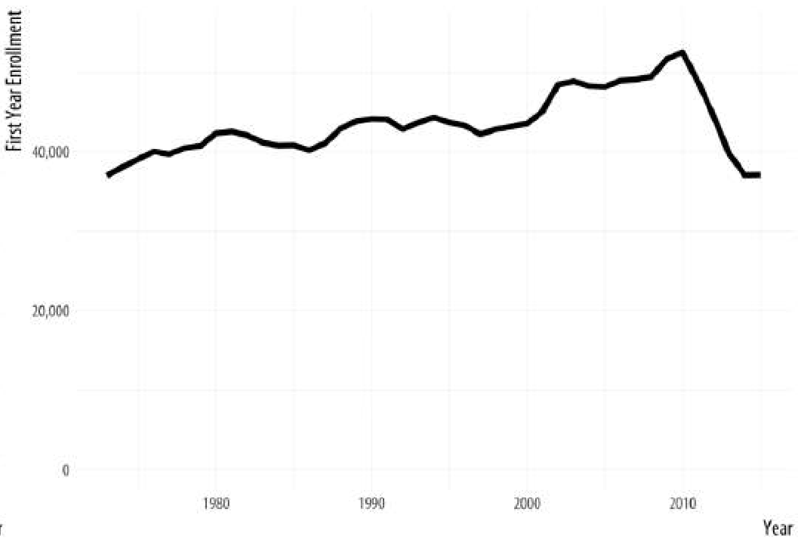
Source: Fox News (2013)

Include the Baseline

The same principle applies to line charts, but with a bit more flexibility.

- Can sometimes exclude zero if we are analyzing subtle variations or trends.
- But if omitting zero distorts the perceived magnitude, the plot becomes misleading.

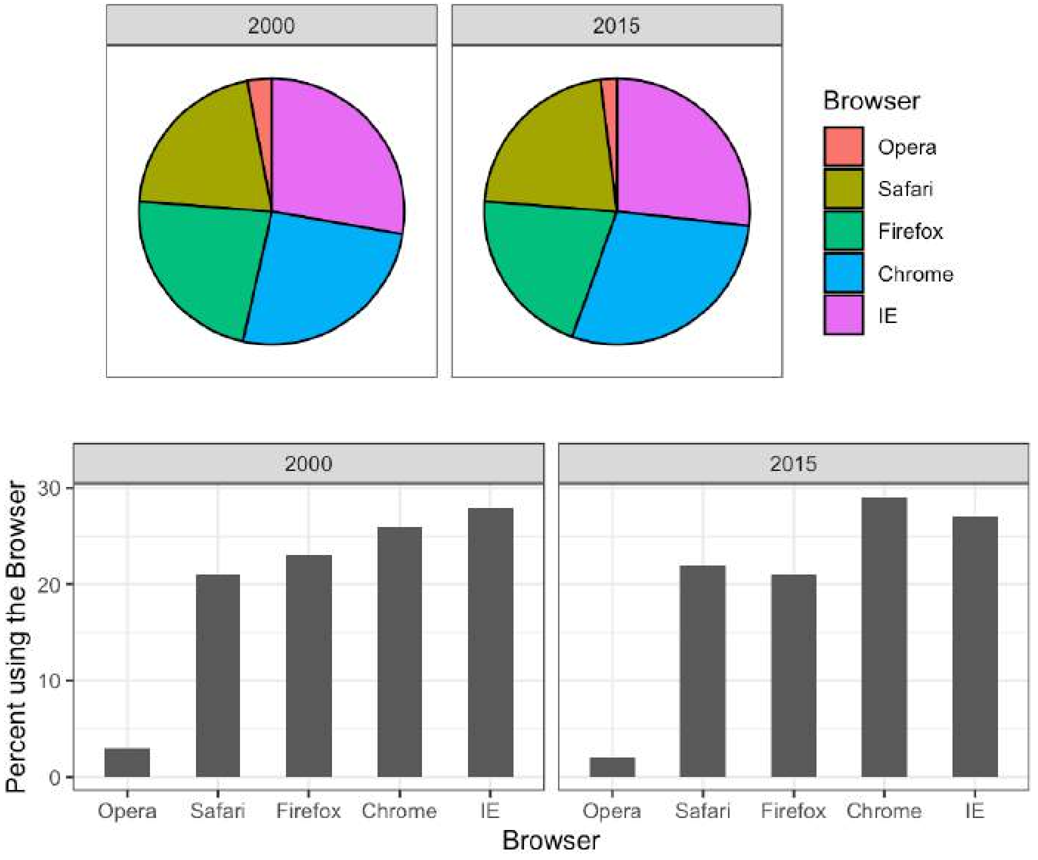




Avoid Using Area

We are much better at judging position and length than area.

- In general, use bars or lines instead of pies.
- Position and length are easier to compare than angles, diameters, or areas.

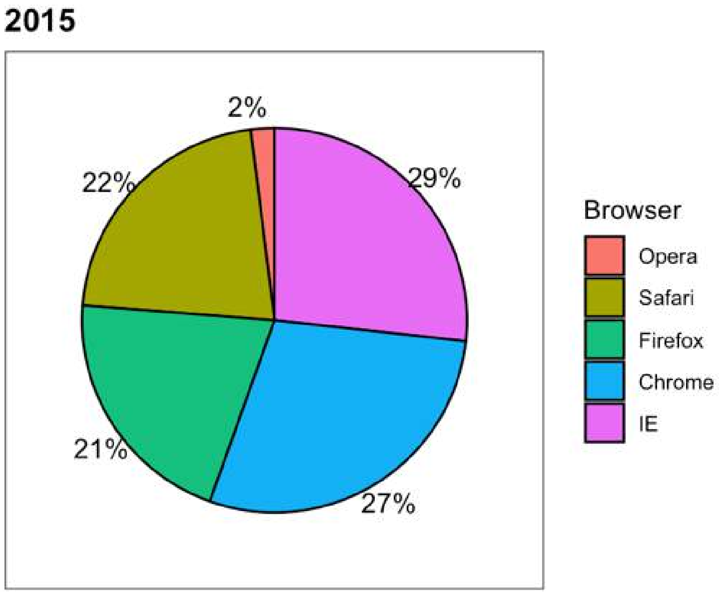


Avoid Using Area

If, for some reason, you must create a pie chart, make it easier to read.

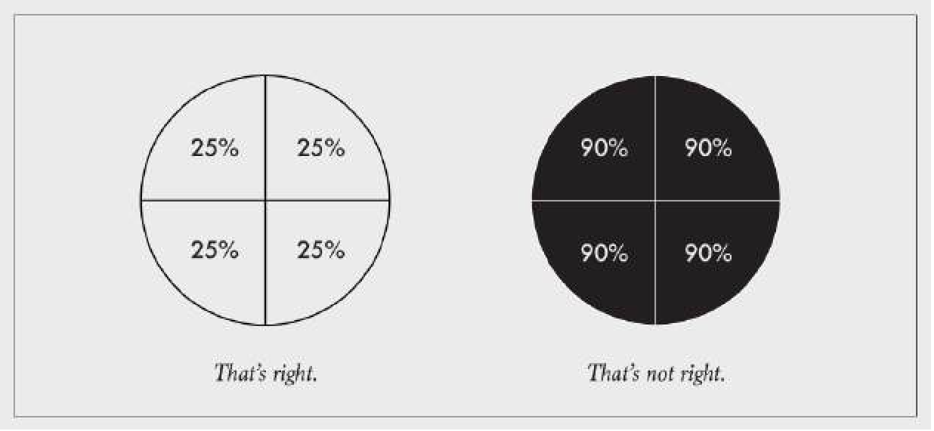
- Label each slice with its respective percentage.
- So your viewer do not have to estimate proportions from angles or area.

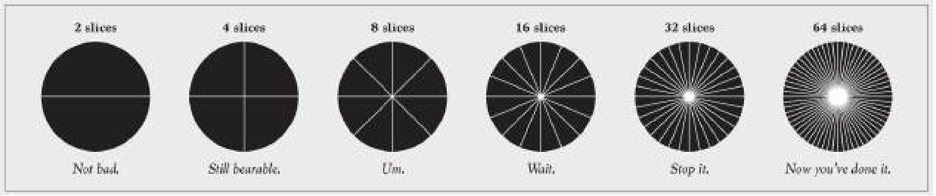
Browser	2000	2015
Opera	3	2
Safari	21	22
Firefox	23	21
Chrome	26	29
IE	28	27



More on Pie Slices

- Pie charts are tolerable with a few categories.
- But quickly become unreadable as slices increase.
- When there are too many small parts, switch to a bar plot.



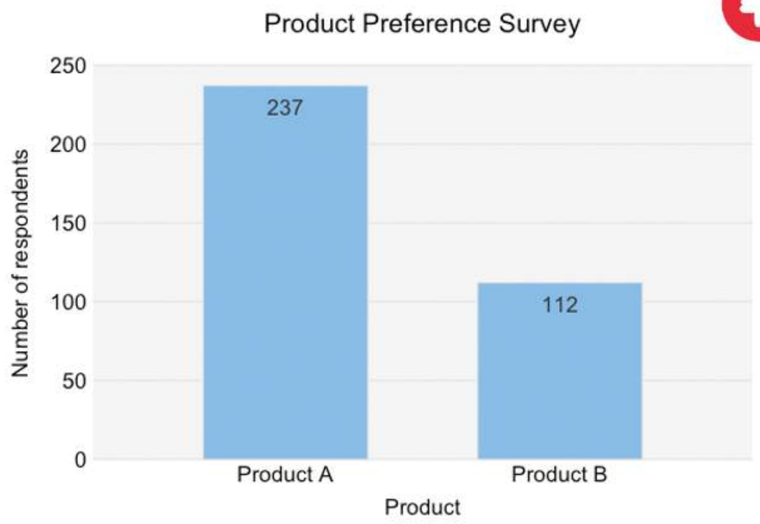


Source: Flowing Data, Nathan Yau

Data Density

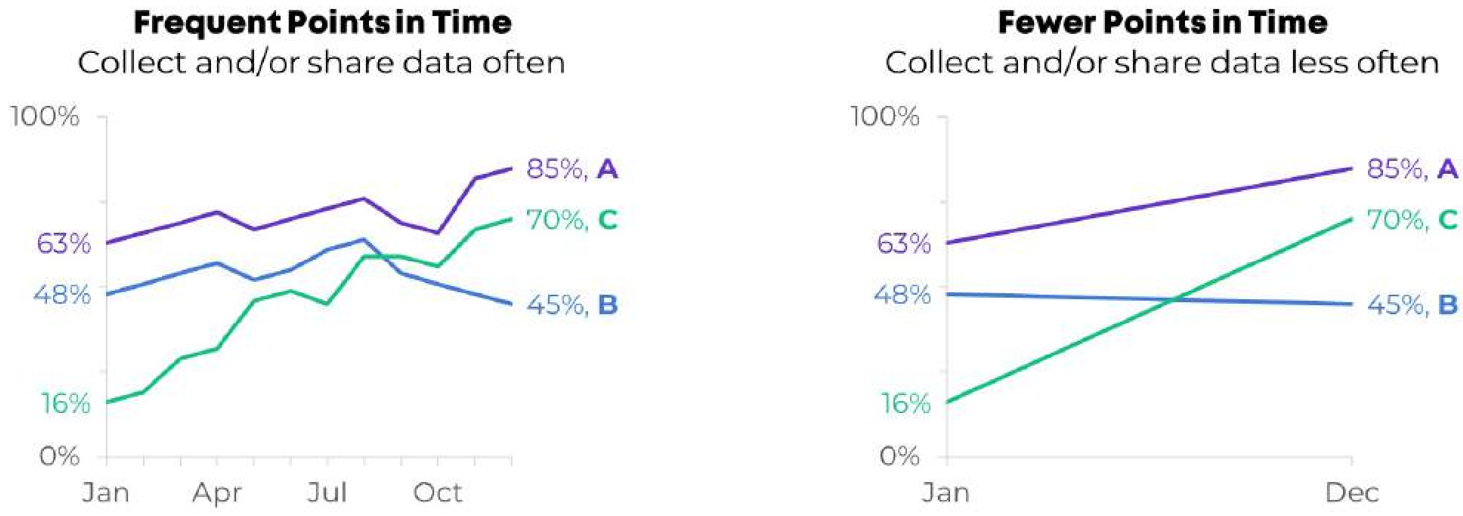
Don’t use graphs to display small amount of data.

- When there is not much to show, a sentence is often better than a chart.
- Your plot should add value, not just take up space.



237 of the respondents preferred Product A, while only 112 preferred Product B.

Consider Your Audience

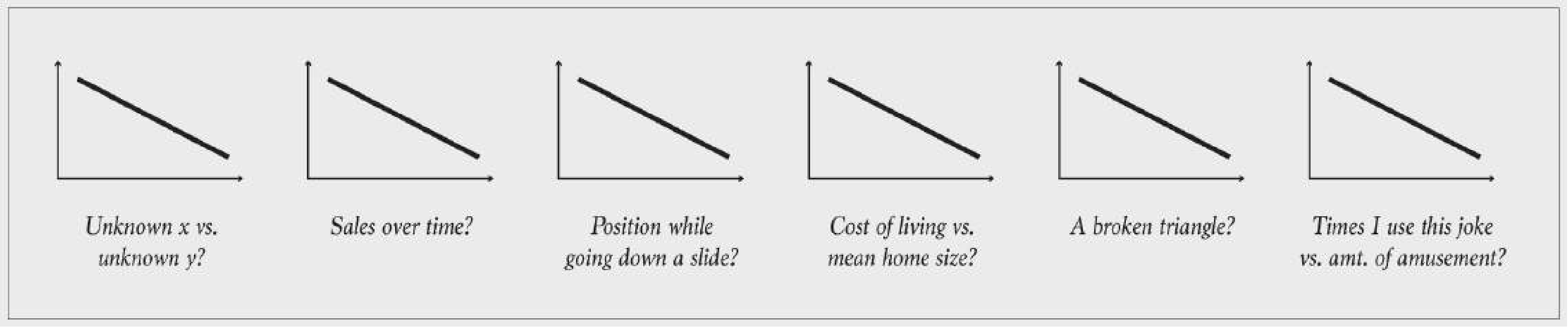


- Your audience likely have different levels of technical expertise, familiarity or the subject, or goals.
- Tailor the amount of detail to what they can meaningfully interpret.

Explain Encodings

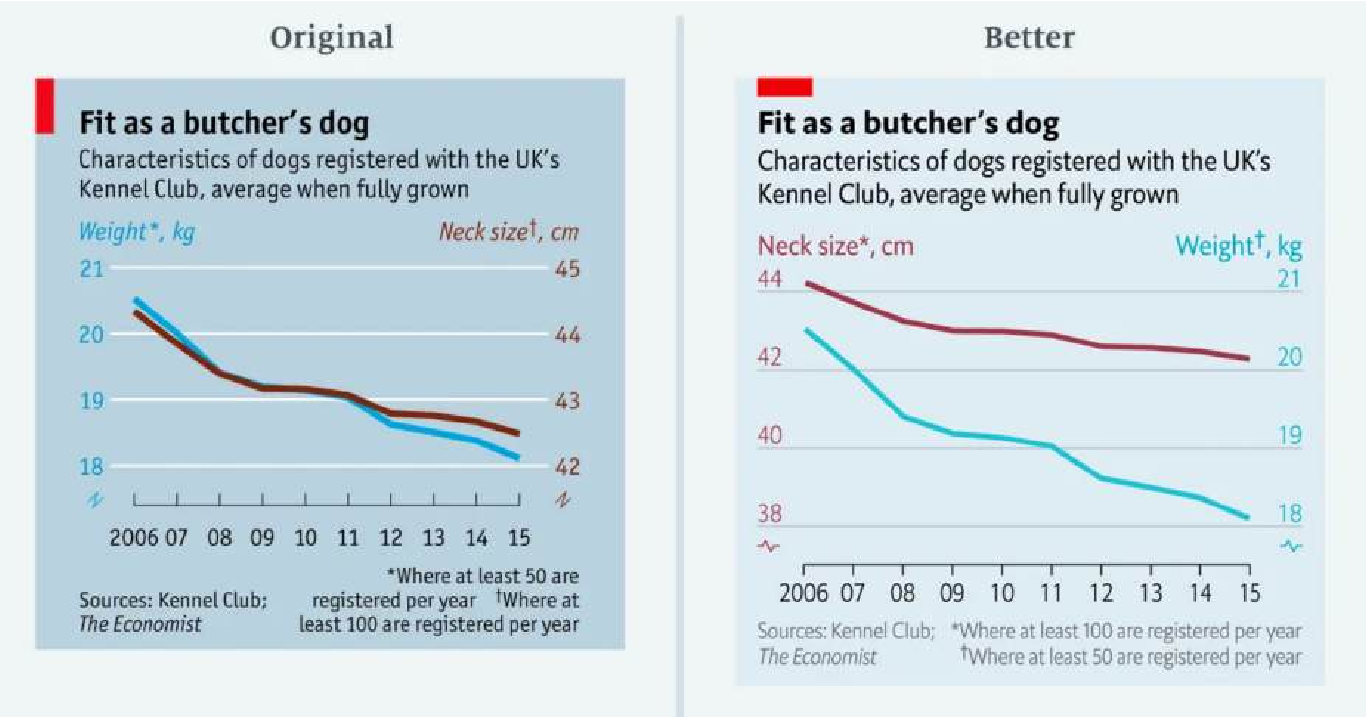
We encode information through shapes, colors, and positions.

- Viewer must be able to **decode** them back to the values.
- Explain the data with title, labels, legends, and annotations.
- Without context, even a simple line can mean many things.



Source: Flowing Data, Nathan Yau

Avoid Cherry-Picking



Cherry-picking can mislead your viewers into believing a trend exists (or not).

- If you use dual axes, make sure the scales represent comparable proportional change.
- If you need to drop values or observations, clearly explain why.

Source: Medium

Preattentive Attributes

How many 9s are there?

8	3	5	7	3	4	8	9	4	4	0	0	7
0	5	2	7	8	5	3	8	4	8	5	1	6
4	1	4	7	1	5	0	2	4	6	3	7	1
8	9	8	3	6	5	4	5	6	8	6	0	7
4	8	7	8	5	0	7	7	6	8	2	8	3
7	7	7	5	7	8	4	7	8	2	6	9	6
3	9	4	2	3	8	0	5	7	5	5	2	6
6	8	0	7	2	0	2	4	5	6	4	7	0
2	5	3	0	1	1	4	0	5	1	1	7	6
1	0	2	0	6	2	8	4	5	9	6	4	8
2	5	3	5	4	3	4	1	5	0	7	2	0
4	8	3	8	1	2	5	0	6	5	8	4	2
9	7	5	5	2	1	5	0	8	3	2	2	5

Preattentive Attributes

How many 9s are there?

8	3	5	7	3	4	8	9	4	4	0	0	7
0	5	2	7	8	5	3	8	4	8	5	1	6
4	1	4	7	1	5	0	2	4	6	3	7	1
8	9	8	3	6	5	4	5	6	8	6	0	7
4	8	7	8	5	0	7	7	6	8	2	8	3
7	7	7	5	7	8	4	7	8	2	6	9	6
3	9	4	2	3	8	0	5	7	5	5	2	6
6	8	0	7	2	0	2	4	5	6	4	7	0
2	5	3	0	1	1	4	0	5	1	1	7	6
1	0	2	0	6	2	8	4	5	9	6	4	8
2	5	3	5	4	3	4	1	5	0	7	2	0
4	8	3	8	1	2	5	0	6	5	8	4	2
9	7	5	5	2	1	5	0	8	3	2	2	5

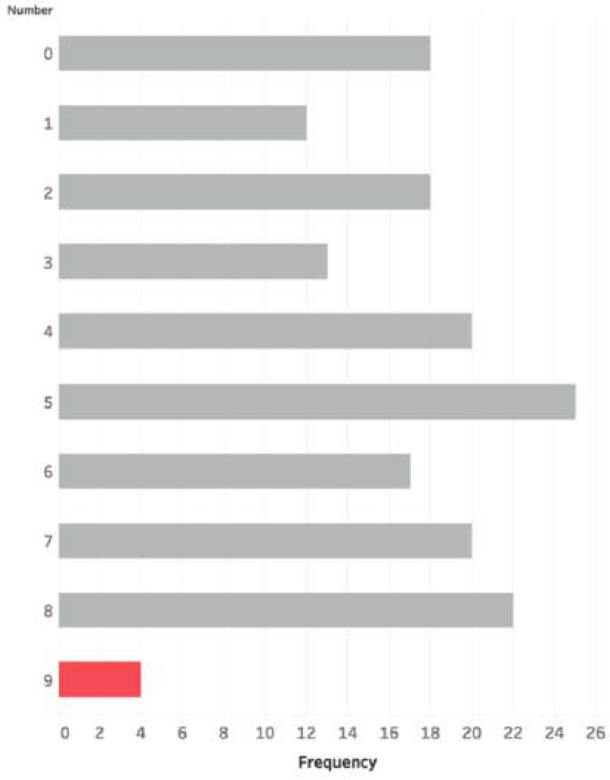
Preattentive Attributes

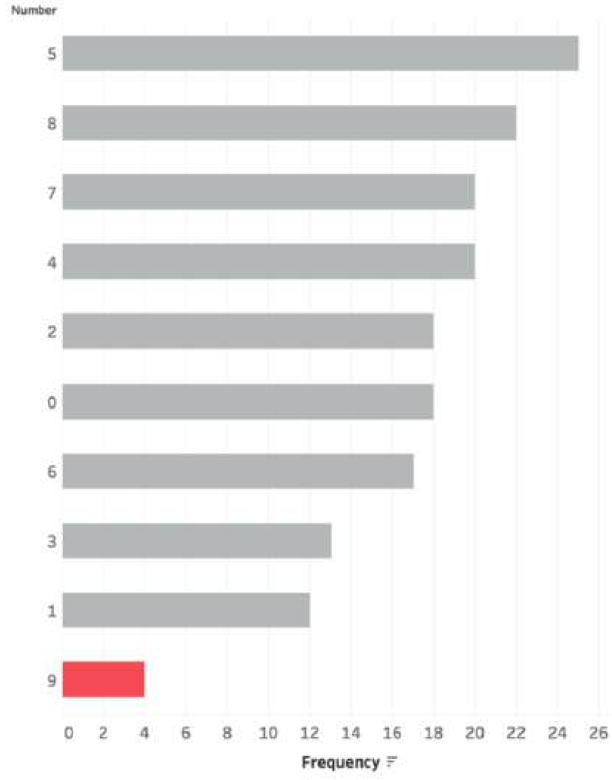
How many 9s are there?

8	3	5	7	3	4	8	9	4	4	0	0	7
0	5	2	7	8	5	3	8	4	8	5	1	6
4	1	4	7	1	5	0	2	4	6	3	7	1
8	9	8	3	6	5	4	5	6	8	6	0	7
4	8	7	8	5	0	7	7	6	8	2	8	3
7	7	7	5	7	8	4	7	8	2	6	9	6
3	9	4	2	3	8	0	5	7	5	5	2	6
6	8	0	7	2	0	2	4	5	6	4	7	0
2	5	3	0	1	1	4	0	5	1	1	7	6
1	0	2	0	6	2	8	4	5	9	6	4	8
2	5	3	5	4	3	4	1	5	0	7	2	0
4	8	3	8	1	2	5	0	6	5	8	4	2
9	7	5	5	2	1	5	0	8	3	2	2	5

Preattentive Attributes

How many 9s are there?





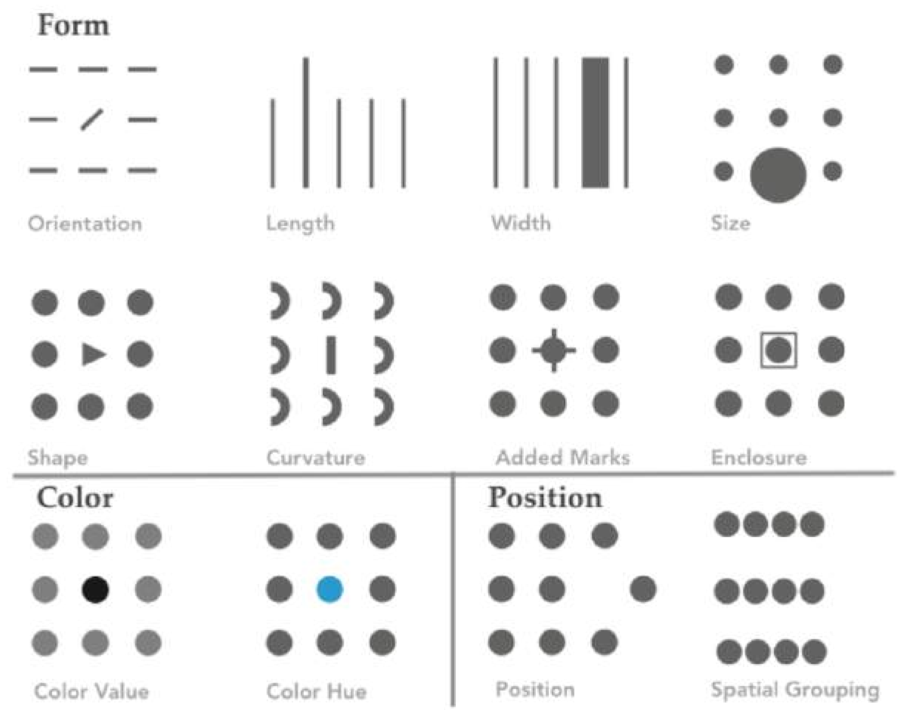
Pre-Attentive Pop-Outs

Some objects in our visual field are easier to see than others.

The examples we just saw used

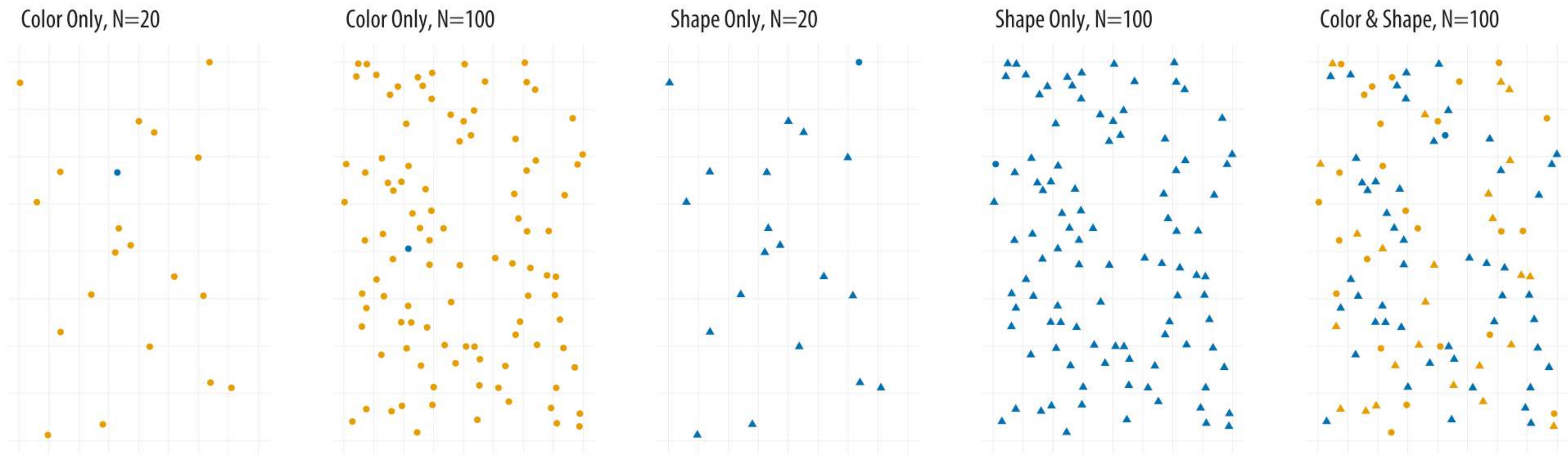
- **color** (9s in red),
- **size** (huge 9s),
- **length** (bar chart), and
- **orientation** (sorted bar chart)

to highlight the 9s.

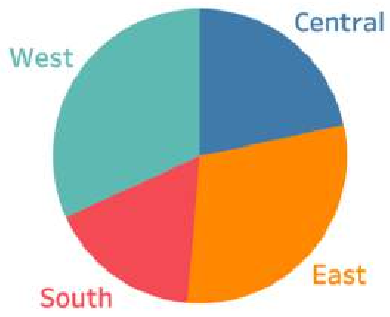


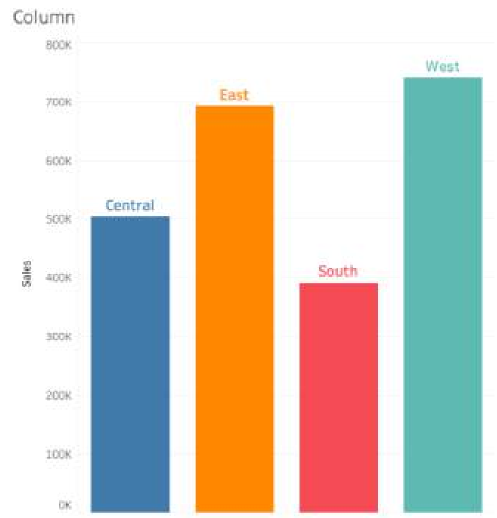
Pre-Attentive Pop-Outs

Pop-outs on the **color channel** is stronger than the shape channel.



Pre-Attentive Pop-Outs

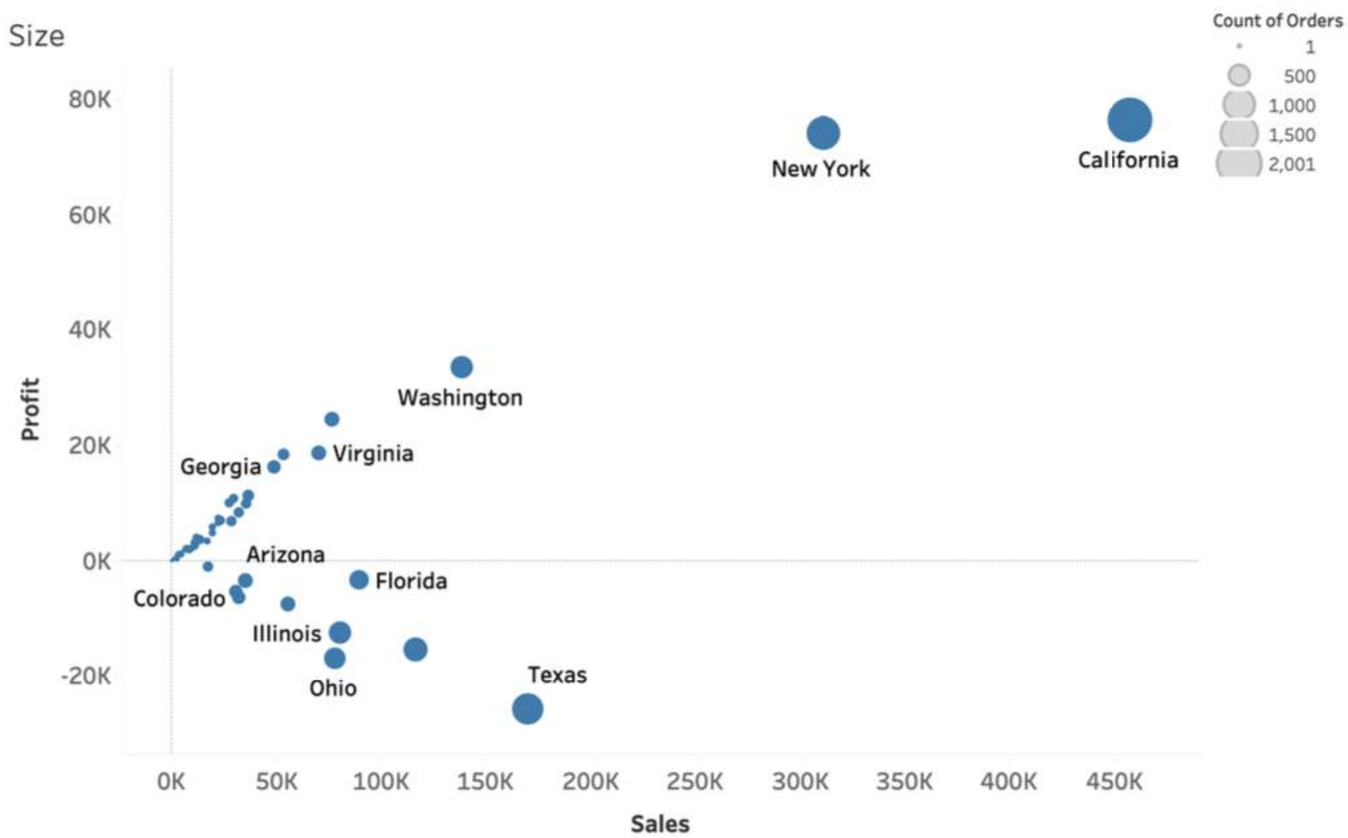




Can you tell which region has the highest sales?

- **Length** is one of the most effective pre-attentive attributes for conveying quantitative information.
- Easy for viewers to compare values quickly.

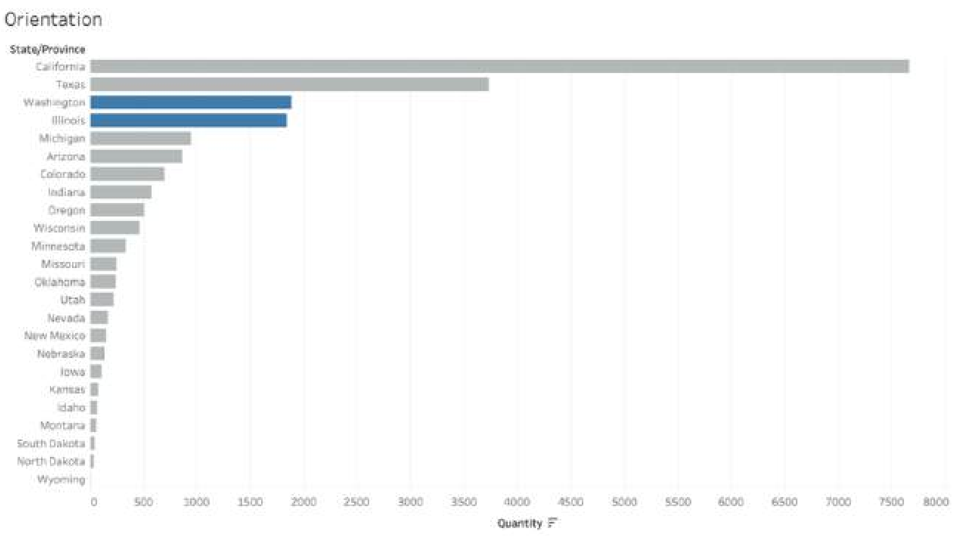
Pre-Attentive Pop-Outs

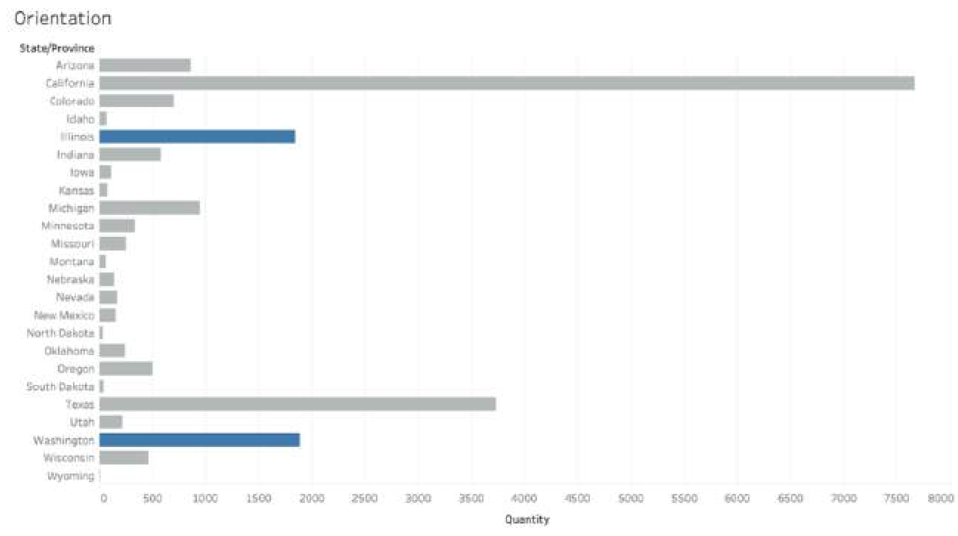


- **Size** emphasises the significance of data points or categories.
- Large sizes draw immediate attention and convey magnitude or importance.

Pre-Attentive Pop-Outs

Can you tell whether Illinois or Washington has a higher transaction quantity?





- **Orientation** is another important pre-attentive attribute in data visualisation.
- It represents trends, directions, and relationships within the data.

Colors

Color is crucial in data visualisation, but it is often misused.

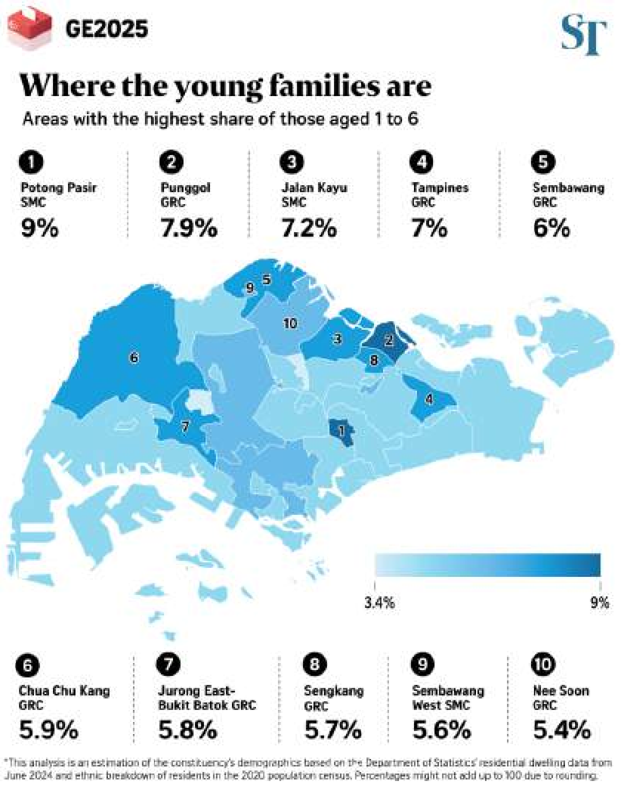
- Instead of decorative, it should serve a specific purpose in enhancing data comprehension.
- Types of color schemes:
  - Sequential: show continuous values.
  - Diverging: emphasise deviation from a midpoint.
  - Categorical: distinguish groups.

Colors

Sequential Color Schemes

Sequential color is the use of a single color that varies in lightness or intensity.

- Typically from light to dark.
- Best for showing continuous data - darker shade represent higher values.

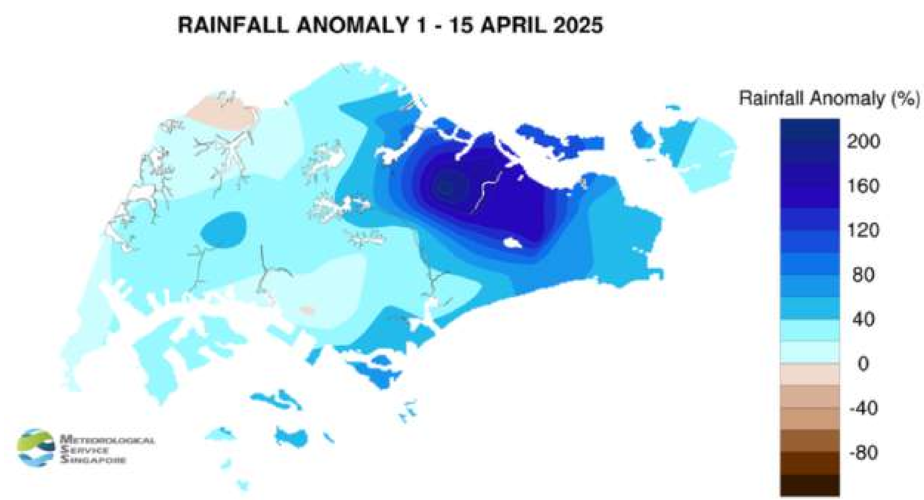


Colors

Diverging Color Schemes

Diverging color is used to show values that deviate around a midpoint.

- Two contrasting colors that move in opposite directions.
- Helps viewers see both the direction and magnitude of change.

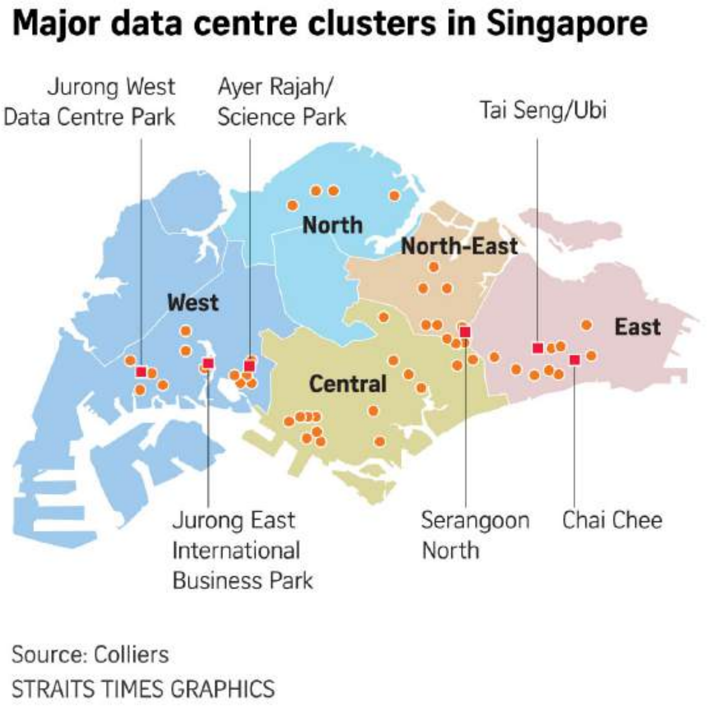


Colors

Categorical Color Schemes

Categorical color uses different colors to distinguish between different groups.

- Each color represents a unique group.
- There is no natural ranking between the groups.

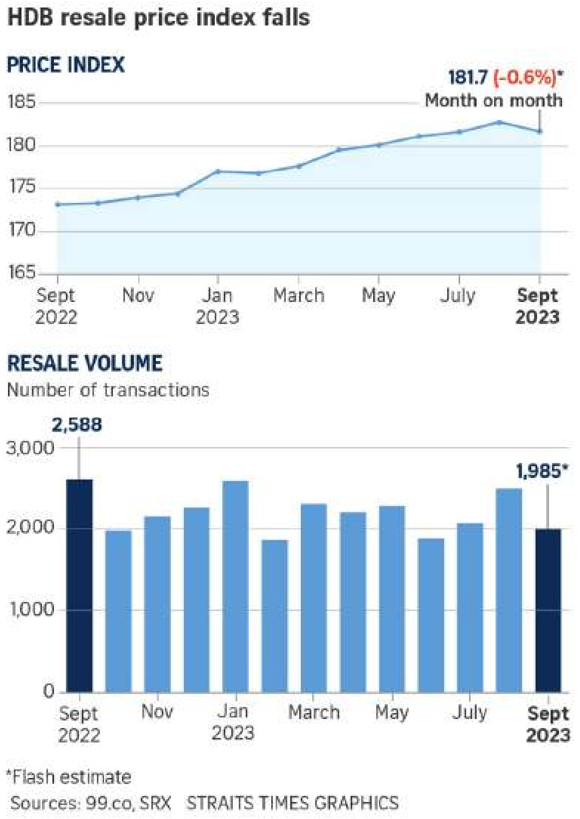


Colors

For Highlighting

Color can be used to **highlight** or **draw attention**.

- Helps viewers quickly spot key patterns and changes.
- Best to limit highlights to one or two key elements, not everywhere.



Colors

Color Psychology and Emotional Associations

People may react differently to colors depending on culture, context, and personal experience.

Color theory provides evidence-based guidance that helps us choose colors that align with meaning and intent in visualisations.

Read more: How color impacts conversion rates and UX?

Color Vision Deficiency (CVD)

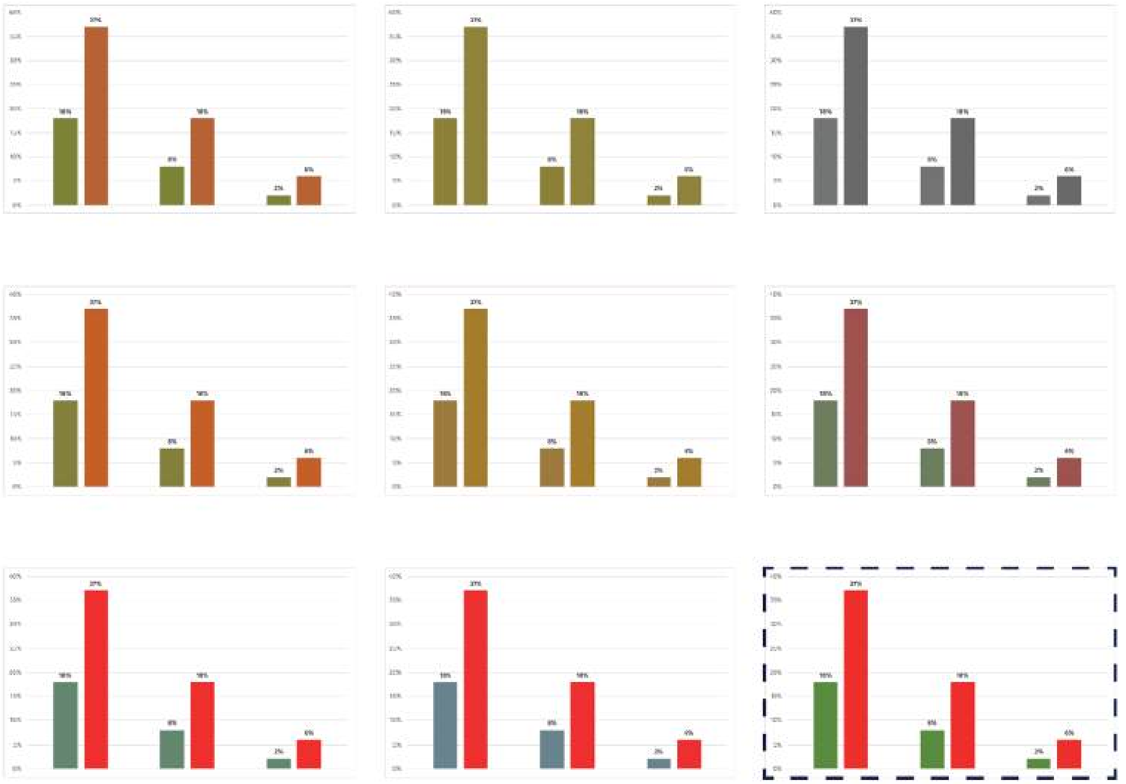


Figure 1: CVD simulation for different types of colour blindness. The original plot is highlighted at the bottom right with a dashed line. First column represents Anomalous Trichromacy (from top to bottom: red-weak, green-weak and blue-weak CVD), middle column represents Dichromatic View (from top to bottom: red-blind, green-blind and blue-blind CVD) and right column represents Monochromatic View (monochromacy and blue cone monochromacy).

Source: Alba Feranandez-Barral (2020)

Color Vision Deficiency (CVD)

Around 1 in 20 people experience some form of color vision deficiency.

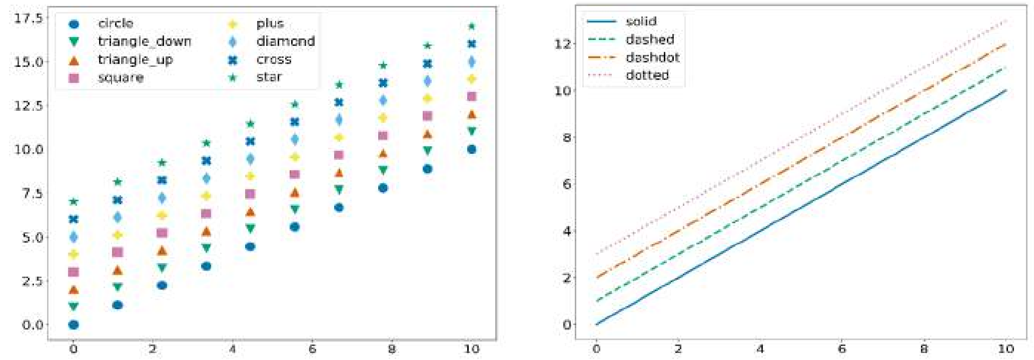


Figure 2. Example of markers (left) and line styles (right) available on Matplotlib. Credit: R. López-Coto (IAA-CSIC).

- Choose a color scheme that can be easily identified by people with all types of color vision such as the Viridis Color Palette.
- Check if your plot is understandable for everybody with color blindness simulators such as Colblindor.
- Avoid relying solely on color. Use line types, shapes, or annotated text to reinforce meaning.
- When communicating, clearly state the color/shape/line type names.

Color Vision Deficiency (CVD)

Support materials

1. Style sheets in `Matplotlib`:

   `tableau-colorblind10` and `seaborn-colorblind10`

2. `Matplotlib` markers:

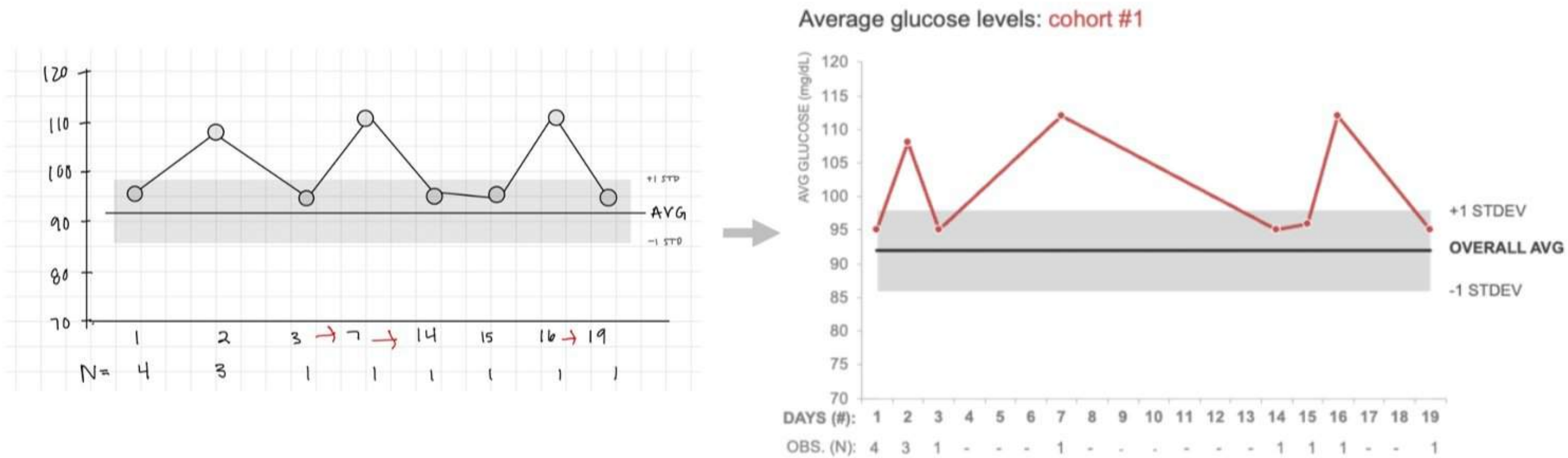
   https://matplotlib.org/3.1.0/api/markers_api.html

3. `Matplotlib` line styles:

   https://matplotlib.org/3.0.3/gallery/lines_bars_and_markers/line_styles_reference.html

Best Practices & References

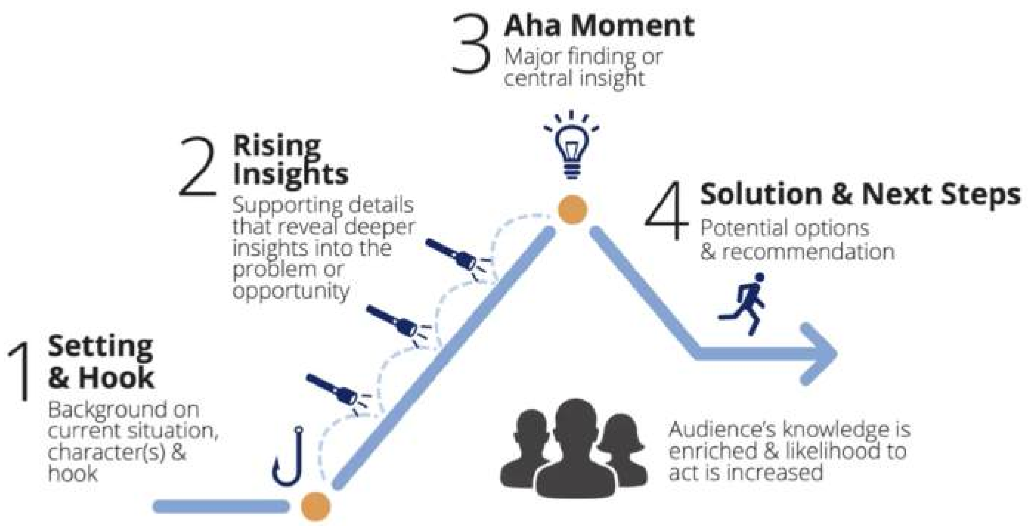
Sketch Your Data



- Some low-tech planning prior to creating visualisations.
- Start with a blank piece of paper, sketch your data to iterate through different approaches.
- Assess what works best, then write code to bring it to life.
- Sketch your data: Bring a table to life.

Source: Cole Nussbaumer Knaflic

The Scientific Process



- Treat each "finding" as a conjecture.
- Do not believe it until you can find multiple ways of confirming it.
- Ask your team members/colleagues to review your work.
- Always follow-up with more questions, investigations, and plots.

Source: Brent Dykes

Summary

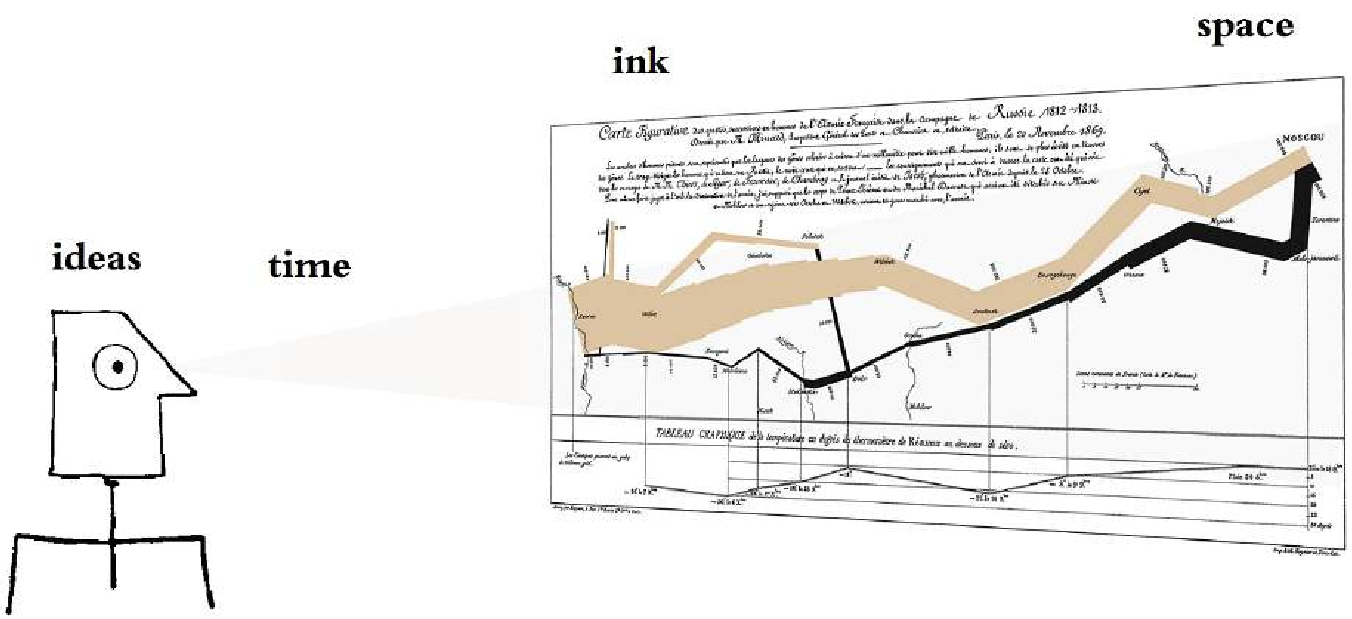
- We learned the basics of visualisation principles for effective communication and data visualisation.
- They acts as the foundation for understanding modern visualisation libraries.
- There are much more to data visualisation than we can possibly cover in this course.

References

Some useful resources if you are interested in exploring more:

- Books by Edward Tufte
  - The Visual Display of Quantitative Information
  - Envisioning Information
  - Visual Explanations
  - Beautiful Evidence
  - Seeing with Fresh Eyes
- Books by Alberto Cairo
  - The Functional Art
  - The Truthful Art
  - How Charts Lie
  - The Art of Insight
- Data Visualization: A Practical Introduction by Kieran Healy
- Fundamentals of Data Visualization by Claus O. Wilke
  - The *Directory of Visualizations* provides a quick visual overview of the various plots and charts that are commonly used to visualize data.
- FlowingData.com by Nathan Yau
  - The *Guides* section is extremely useful for practitioners.
- Storytelling with Data by Cole Nussbaumer Knaflic
- Effective Data Storytelling by Brent Dykes

Epilogue



A figure presented by Edward Tufte, who said

> “Graphical excellence is that which gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space.”